

Math 250A Examples.

These are a rough collection of notes of examples taken from Professor Richard E. Borcherds' Math 250A lectures during the Fall 2025 semester. Please send an email to nydhruva@berkeley.edu if you see any misprints (there are probably quite a few).

1. **Cayley graph ex.** For a set S of 5 points, let $G = \mathbb{Z}/5\mathbb{Z}$ and draw the Cayley graph of G acting on S .
2. **permutation rep. ex.** Let $G = D_{20}$ the dihedral group of a icosahedron. We have a transitive representation of G on S . On the icosahedron, we have 12 vertices, 20 faces, and 30 edges. Pairing them up gives 6 pairs of vertices, 10 pairs of faces, and 15 pairs of edges. There exists a permutation representation on all points of the icosahedron.
3. **isomorphism ex.** We have the isomorphism $(\mathbb{Z}/5\mathbb{Z})^* \cong \mathbb{Z}/4\mathbb{Z}$. The elements of $(\mathbb{Z}/5\mathbb{Z})^*$ are $2^0, \dots, 2^3$, which are precisely 1, 2, 4, 3, which can be sent (respectively) to 0, 1, 2, 3.
4. **Classification of groups ex.** Of order 1, it is only the trivial group. Of order 2, it is $\mathbb{Z}/2\mathbb{Z}$. Of order 3, it is $\mathbb{Z}/3\mathbb{Z}$. Of order p (where p is prime), we can pick $g \in G$. By Lagrange's theorem, the order of g divides $|G| = p$, so g has order 1 or p . Thus, $G \cong \mathbb{Z}/p\mathbb{Z}$. Of order 4, every element has order 1, 2, or 4. By a Lang exercise, the group of order 4 must be abelian.
5. **rotations of icosahedron ex.** Let S be the set of vertices of an icosahedron. Then $|S| = 12$. The subgroup fixing vertex is $\mathbb{Z}/5\mathbb{Z}$. Thus $|G| = 12 \times 5 = 60$.
6. **group product ex.** $\mathbb{Z}/6\mathbb{Z} \cong \mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$. Additionally, all symmetries of the icosahedron (including reflections) form a group of order 120 and splits as a product $\mathbb{Z}/2\mathbb{Z} \times (\text{rotations})$, where we have that vertex can be sent to its "opposite" vertex. Similar logic holds for the cube, octahedron, and dodecahedron. However, it fails for the tetradron, as there are no "opposite" vertices. We may also look at the group G of roots of unity, which are union of roots of order n , where $\mathbb{Z}/n\mathbb{Z} = \mathbb{Z}/a\mathbb{Z} \times \mathbb{Z}/b\mathbb{Z}$, where $n = ab$ with $\gcd(a, b) = 1$, and group G can be written as the product

$$(\text{roots of odd order}) \times (\text{roots of order a power of 2})$$

Lastly, the group $(\mathbb{R}, +)$ is a group that is a product of groups. $\mathbb{R}$ is a vector space over $\mathbb{Q}$ so it has a basis S . Writing S as a union $S_1 \cup S_2$ gives $\mathbb{R} = (\text{vector space basis } S_1) \times (\text{vector space basis } S_2)$.

7. **left cosets $\neq$ right cosets.** Take $G = S_3 = \{e, (12), (23), (13), (123), (132)\}$. Let $H = \{e, (12)\}$ be a subgroup of G . The left cosets are $H, \{(13), (123)\}, \{(23), (132)\}$, while the right cosets are $H, \{(132), (13)\}, \{(23), (123)\}$, which are different cosets.
8. **exact sequence, (non)uniquely determined group.** Suppose A, C are groups and we have an exact sequence $1 \rightarrow A \rightarrow B \rightarrow C \rightarrow 1$. Can we uniquely determine B ? Counterexamples include $0 \rightarrow \mathbb{Z} \rightarrow \mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z} \rightarrow 0$ and $0 \rightarrow \mathbb{Z} \rightarrow \mathbb{Z} \times \mathbb{Z}/2\mathbb{Z} \rightarrow \mathbb{Z} \rightarrow 0$.

9. **classify groups of order 8.** In case 1, suppose all elements have order 1 or 2. Then $G = (\mathbb{Z}/2\mathbb{Z})^3$. In case 2, suppose G has element a of order 4. Put $H = \langle a \rangle$. Then we have $1 \rightarrow \mathbb{Z}/4\mathbb{Z} \rightarrow G \rightarrow \mathbb{Z}/2\mathbb{Z} \rightarrow 1$. Then $\mathbb{Z}/2\mathbb{Z}$ either acts on $\mathbb{Z}/4\mathbb{Z}$ trivially or by having $1 \in \mathbb{Z}/2\mathbb{Z}$ send $a \in \mathbb{Z}/4\mathbb{Z}$ to $-a$. We then have a table.

placeholder	trivial action	nontrivial action
$b^2 = 1$	$\mathbb{Z}/4\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$	$\mathbb{Z}/4\mathbb{Z} \rtimes \mathbb{Z}/2\mathbb{Z}$
$b^2 = a$	$b^8 = 1$ same as $\mathbb{Z}/8\mathbb{Z}$	DNE
$b^2 = a^2$	$c = ba, c^2 = e, a^4 = 1, ca = ac$	$a^4 = 1, b^2 = a^2, bab^{-1} = a^{-1}$ same as quaternion group

10. **SO3 example.** Then S^3 acts on $\mathbb{R}^3$ by $g(bi + cj + dk) = g(bi + cj + dk)g^{-1}$, where the LHS is the group action and the RHS is quaternion multiplication. This gives a homomorphism from S^3 onto SO3, with kernel $\{-1, 1\}$. The only spheres that are groups are S^0 (related to $\mathbb{R}$), S^1 (related to $\mathbb{C}$), and S^3 (related to $\mathbb{H}$, the quaternions). In matrix form, the quaternions can be thought of as generated by the following 4 matrices:

$$\begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

where $I = \sqrt{-1}$ in $\mathbb{C}$.

11. **action by conjugation.** If G is a group, it is sometimes useful to consider when G acts on G by conjugation.
12. **Dihedral group of square.** List the elements of the dihedral group of a square and try to pair them up (as much as you can) into pairs that have the same number of fixed points. Also, for each symmetry of the square, try to find the number of conjugate elements.
13. **[COUNTEREXAMPLE] two elements conjugate.** Here we list a counterexample to the claim: any two elements in a group are conjugate. Consider a pentagon. Let g_1 be a “single” rotation of the pentagon clockwise and g_2 be similarly defined but anticlockwise. Then g_1 and g_2 are conjugate in D_{10} (the dihedral group of order 10) but are not conjugate in $\mathbb{Z}/5\mathbb{Z}$.
14. **Example of nilpotent group.** Let G be the subgroup of $GL_n(\mathbb{Z}/p\mathbb{Z})$ so that elements of G are upper-triangular matrices with 1’s on the diagonal. Then $Z(G)$ is a nilpotent group.
15. **Only nilpotent group of order less than 10.** The only nilpotent group of order < 10 is S_3 .
16. **Dihedral groups are symmetries of the regular n -gon.** Make and fill a table with columns: n -value, corresponding dihedral group order, what the

dihedral group is isomorphic to, how many conjugacy classes are there of order 2, and the order of the center of the respective dihedral group.

17. **Dihedral group isomorphism.** Show that $D_{12} \simeq D_6 \times \mathbb{Z}/2\mathbb{Z}$.
18. **D_6 example.** Draw a hexagon. Draw two triangles (call them yellow and blue) inside it that are “tilted” versions of each other. D_6 is the symmetries of the yellow triangle and $\mathbb{Z}/2\mathbb{Z}$ are 180-degree rotations. However, Hexagon rotation Prop. does not work for D_{16} .
19. **Binary dihedral group.** This is the “forgotten” group of order 12, which is the binary dihedral group, which is the semidirect product $\mathbb{Z}/3\mathbb{Z} \rtimes \mathbb{Z}/4\mathbb{Z}$. Looking at the semidirect product, we get that $\mathbb{Z}/3\mathbb{Z}$ acts nontrivially on $\mathbb{Z}/4\mathbb{Z}$ as an element of order 2 in $\mathbb{Z}/4\mathbb{Z}$ acts trivially on $\mathbb{Z}/3\mathbb{Z}$. We get the binary dihedral group by looking at $S^3 \xrightarrow{\text{onto}} SO3 \rightarrow 1$. We get that S_3 is a subgroup of $SO3$ (S_3 is dihedral) and put $\hat{S}_3$ as the inverse image of S_3 in S^3 .
20. **[COUNTEREXAMPLE-ish] to Sylow Theorem 2.** Sylow Theorem 2 fails for order p , since we can take D_8 and see that it has 5 subgroups. We have that the element of D_8 taking bottom-left vertex to top-right vertex (and vice-versa) is not conjugate to the element of D_8 taking top edge to bottom edge (and vice-versa). D_8 has 3 subgroups of order 4. Also, the order 4 subgroups ($\mathbb{Z}/4\mathbb{Z}$) are not isomorphic to the group generated by the elements g, h where g takes bottom edge to top edge (and vice-versa) and h takes right edge to left edge (and vice-versa), which is the group $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$.
21. **Application of Sylow Prop.** We look at order 12 dihedral group G . Look at Sylow 3-subgroups (order 3). There should 4 Sylow 3-subgroups, where each of them has 2 elements of order 3, disjoint (apart from). So G has 8 elements of order 3. $12-8=4$ elements of order $\neq 3$. Also these elements of order $\neq 3$ form a Sylow subgroup, which must be these 4 elements, so this Sylow subgroup of 4 elements must be normal.
22. **Classification of order 12 groups into the number of normal sylow subgroups.** Make a column for the number of sylow 2-subgroups and one for the number of sylow 3-subgroups. The possibilities are yes/yes, yes/no, and no/yes. For the first case, we have 2 abelian groups which are $\mathbb{Z}/3\mathbb{Z} \times (\mathbb{Z}/2\mathbb{Z})^2$ and $\mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z}$. For the second case, we have a semidirect product $\mathbb{Z}/4\mathbb{Z} \rtimes \mathbb{Z}/3\mathbb{Z} \simeq A_4$. For the third case, we have $\mathbb{Z}/3\mathbb{Z} \rtimes (\mathbb{Z}/2\mathbb{Z})^2 \simeq D_{12} \simeq D_6 \times \mathbb{Z}/2\mathbb{Z}$ and $\mathbb{Z}/3\mathbb{Z} \rtimes \mathbb{Z}/4\mathbb{Z}$, which is the binary dihedral group.
23. **order 13, 14, 15 groups.** For order 13, since 13 is prime, we have only $\mathbb{Z}/13\mathbb{Z}$. For order 14, it is either cyclic or dihedral. For order 15, realize $15 = p \cdot q = 3 \cdot 5$, where p, q are primes and $p < q$. The number of subgroups of order q is 1 (mod q) and divides pq so must be 1.
24. **order 16 groups.** Possibilities are:
 - (a) abelian groups (5 possibilities)

- (b) 4 more with cyclic subgroup of order 8
 - (c) 2 products (order 2) $\times$ (nonabelian of order 8)
 - (d) 2 semidirect products (order 4) $\rtimes$ (cyclic of order 4 acting nontrivially)
 - (e) semidirect product $(\mathbb{Z}/4\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}) \rtimes \mathbb{Z}/2\mathbb{Z}$.
25. **Counting number of homomorphisms.** Let $G \simeq (\mathbb{Z}/p\mathbb{Z})^a \times (\mathbb{Z}/p^2\mathbb{Z})^b$. We count the number of homomorphisms $\phi : G \rightarrow \mathbb{Z}/p\mathbb{Z}$, which is p^{a+b} many homomorphisms. The number of homomorphisms $\psi : G \rightarrow \mathbb{Z}/p^2\mathbb{Z}$ is p^{a+2b} . Then we have that the knowledge of $a + b$ and $a + 2b$ determines the values of a and b .
26. **Abelian groups of order 1,000,000.** Notice $1,000,000 = 2^6 \cdot 5^6$. Now look at the number of ways of writing 6 as a sum: $6 = 6 + 0 = 4 + 2 = 4 + 1 + 1 = 3 + 2 + 1 = 3 + 3 = 3 + 1 + 1 + 1 = 2 + 2 + 2 = 2 + 2 + 1 + 1 = 2 + 1 + 1 + 1 + 1 = 1 + 1 + 1 + 1 + 1 + 1$. Thus, there are 11 ways of writing 6 as a sum. Thus, by the fundamental theorem of finite abelian groups, there are 11 abelian groups of order 2^6 . Similarly, there are 11 abelian groups of order 5^6 . Thus, there are $11 \times 11 = 121$ abelian groups of order $10^6 = 1,000,000$.
27. **groups of order 17, 18, 19, 20, 21, 22, 23.** 17, 19, and 23 are prime, so they each just give one group, $\mathbb{Z}/p\mathbb{Z}$. For order 18, there are 5 groups: (order 9) $\rtimes$ (order 2). For order 20: $\mathbb{Z}/5\mathbb{Z} \rtimes$ (order 4). For order 21 we have that $21 = p \cdot q$, so we get $\mathbb{Z}/7\mathbb{Z} \rtimes \mathbb{Z}/3\mathbb{Z}$, which is the first nonabelian group of odd order (in total there are 2 groups of order 21). For order 22, similar case to order 21 groups and we get 2 groups of order 22.
28. **wreath product example.** Take $\mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$. We have that the group $\mathbb{Z}/2\mathbb{Z}$ acts on this by swapping the copies. So we get a wreath product $(\mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}) \rtimes \mathbb{Z}/2\mathbb{Z}$.
29. **more wreath product example.** Sylow subgroups of symmetric groups S_n . Take 10 points in a row and put the points in a set S . Then we can pair up the points so that $\mathbb{Z}/2\mathbb{Z}$ acts on each pair. There are 5 pairs, so thus $(\mathbb{Z}/2\mathbb{Z})^2$ (of order 2^5) acts on the set S . Then, we can swap pairs, and consider the other ways of swapping pairs of pairs, etc. This gives wreath products $(\mathbb{Z}/2\mathbb{Z} \wr \mathbb{Z}/2\mathbb{Z}) \times \mathbb{Z}/2\mathbb{Z}$ and $(\mathbb{Z}/2\mathbb{Z} \wr \mathbb{Z}/2\mathbb{Z}) \wr \mathbb{Z}/2\mathbb{Z}$.
30. **another wreath product example (symmetry group of tree).** Draw a tree as follows. Draw a big “+” sign. Then for each of the four tips, draw 3 “symmetric” lines coming out. Then the symmetry group of this tree is $(S_3) \wr S_4$.
31. **Order 24 groups.** Look at Sylow 3-subgroups. The number of them divides 24 and is $\equiv 1 \pmod{3}$. So, there must be either 1 or 4 Sylow 3-subgroups. If only 1, then it is normal and we get $G = (\text{syLOW 3-subgroup}) \rtimes (\text{subgroup of order 8})$. If there are 4 sylow 3-subgroups, then G acts transitively on these by conjugation. Then there is a homomorphism $\phi : G \rightarrow S_4$. $\text{image}\phi$ is divisible by 4 and $\ker \phi$ has order 1,2,3, or 6 but $\ker \phi$ can't have order 3 or 6 since G

has a normal subgroup of order 3. If $|\ker \phi| = 1$, then $G = S_4$. If $|\ker \phi| = 2$, then $1 \rightarrow \mathbb{Z}/2\mathbb{Z} \rightarrow G \rightarrow A_4 \rightarrow 1$. Possibilities include S_4 , $A_4 \times \mathbb{Z}/2\mathbb{Z}$, and $\hat{A}_4 = \mathbb{Z}/2\mathbb{Z} \rtimes A_4$.

32. **Normal subgroups of symmetric groups.** Does S_n have normal subgroups other than 1, S_n , A_n ? Note S_4 has a normal subgroup isomorphic to $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$. S_4 can be viewed as the rotations of a cube. so we get a homomorphism $\phi : S_4 \rightarrow S_3$ that is onto with kernel that is a normal subgroup of order 4.
33. **Find c .** Now find an element $c \in S_n$ so that $c^{-1}ac = b$ for a given $a, b \in S_n$. To do this, write a, b in their cyclic decomposition form. Since they have the same cyclic shape (by assumption), thus “align” their respective cycles together and define c taking the corresponding elements of a to the corresponding elements of b .
34. **order 48 groups.** Formed out of S_4 and $\mathbb{Z}/2\mathbb{Z}$. We have $GL_2(\mathbb{Z}/3\mathbb{Z})$ (which has order $(3^2 - 1)(3^2 - 3)$). Also we have $\mathbb{Z}/2\mathbb{Z} \times S_4$. Also, we have the binary octahedral group $S_4 \rtimes \mathbb{Z}/2\mathbb{Z}$.
35. **order 60 groups.** A group of order 60 is the group G of rotations icosahedron (nonabelian simple group). Look at the conjugacy classes. 1 rotation defined by 0 radians identity (order 1). 12 rotations by $2\pi/5$ radians (order 5), which are the same as the rotations by $8\pi/5$. 12 rotations by $4\pi/5$ radians (order 5), which are the same as the rotations by $6\pi/5$. 20 rotations by $2\pi/3$ radians. 15 rotations by π radians. Now, show that G is simple. Let H be a normal subgroup of G . Put $n := |H|$. We know $n \mid |G|$ and n is the sum of some of 1, 12, 12, 15, 20, and H is the union of conjugacy classes. Thus the only possible solutions for n are 1 and $1 + 15 + 12 + 12 + 20$, so G is simple.
36. **Simple group example.** Examples are $\mathbb{Z}/p\mathbb{Z}$. A_n for $n \geq 5$. $GL_n(\mathbb{F}_q)$ q prime. However, the GL_n example is not quite simple, since we have a few glitches. These glitches are that $\det : GL_n(\mathbb{F}_q) \rightarrow \mathbb{F}_q^\times$ and its center is the diagonal matrices. If $n = 1$, then $GL_n(\mathbb{F}_q)$ is abelian. NOTE THAT: $SL_n(\mathbb{F}_q)/\text{center}$ (for $n \geq 2$) is usually simple, except for $PSL(F_2), PSL(F_3)$.
37. **Groups formed out $A_5, \mathbb{Z}/2\mathbb{Z}$.** These are $S_5, A_5 \times \mathbb{Z}/2\mathbb{Z}$, and the binary icosahedral group (which is the fundamental group of the Poincare 3-sphere).
38. **Poincare 3-sphere.** Let M be a 3-manifold (compact). The question is: if $H_1(M) = 1$, then is $M \simeq S^3$? The answer is no. The counterexample is the Poincare 3-sphere constructed as follows. Let G be the binary icosahedral group. Then $G \leq S^3$. The Poincare 3-sphere is S^3/G . For S^3/G , there are no nontrivial fixed points and the Poincare 3-sphere is a manifold. Now, we show S^3/G not isomorphic to S^3 . We have $\pi_1(S^3/G) \simeq G \neq 1 = \pi_1(S^3)$. We have that $H_1(S^3/G)$ is the abelianization of G .
39. **Automorphism group COUNTEREXAMPLE.** The following is a counterexample to the claim: $\text{Aut}(S_n) \simeq S_n$. Take S_6 . We show that S_6 has outer

automorphism. S_5 has a subgroup of order 20 (which has index 6). Take S_5 to act on $\mathbb{Z}/5\mathbb{Z}$. All permutations of $\mathbb{Z}/5\mathbb{Z}$ of form $x \mapsto ax + b$ with $a, b \in F_5$ and $a \neq 0$ (the $ax + b$ group). So S_5 acts transitively on 6 points, since S_5 has a subgroup of order 20 which has index 6. So we have a homomorphism $\phi : S_5 \rightarrow S_6$ with transitive image. S_6 has 6 boring S_5 subgroups (boring referring to subgroups fixing a point). But it also has (6) other weird subgroups (weird referring to subgroups not fixing a point). Let H be a weird S_5 subgroup of S_6 . Then H has index 6. Thus, the action of S_6 on 6 points (with the points being the cosets of H) results in a homomorphism $\psi : S_6 \rightarrow S_6$ so that weird $S_5H \mapsto$ boring S_5 fixing point. This can't be an inner automorphism, as inner automorphisms only taking boring S_5 to boring S_5 .

40. **proof that reduced words is isomorphic to a free group.** Define an action of the free group on the set of reduced words. Suppose the free group is generated by a, b . For a reduced word w , define:

- (a) $a(w) = aw$ if $w \neq a^{-1}b$ and $a(w) = v$ if $w = a^{-1}v$.
- (b) $a^{-1}(w) = a^{-1}w$ if $w \neq av$ and $a^{-1}(w) = v$ if $w = av$.

Do similarly for b . It's easy to check that these take reduced words to reduced words. Also, $a(a^{-1}w) = w$ and $a^{-1}(aw) = w$. Thus we have a well-defined action. Define now a map (free group) $\rightarrow$ (reduced words) with $g \mapsto g(\epsilon)$. Also, (reduced words) $\xrightarrow{\text{obviously onto}}$ (free group), and both maps are bijections.

41. **visualizations of free groups.** Let G be a free abelian group on 2 generators. Then it "looks like" $\mathbb{Z}^2$ in Euclidean space $\mathbb{R}^2$. Now let G be a nonabelian free group. It "looks like" hyperbolic space.
42. **number of homs.** Let $G = \mathbb{Z}/2\mathbb{Z}$. Then the number of homs. from free group on n generators to G is 2^n , which determines n .
43. **fundamental groups of graphs.** Draw a 4-petal flower centered at a single vertex. Then the fundamental group of this graph is free on the n generators (where here $n = 4$). Thus, paths $\iff$ words $\iff$. Thus, fundamental group $\iff$ reduced paths $\iff$ reduced words $\iff$ free group. Also, the fundamental group of a graph is unchanged if we take the graph and adjoin it to a "far-away" vertex via a single edge. Also the fundamental group of a graph is unchanged if we have two graphs adjoined to each other by a single "long" vertex.
44. **graph with fundamental group H .** Let H be a subgroup of a free group G . We will construct a graph with fund. group H . $H \iff$ an action of G on n points (transitive). Graph: vertex for each of the points and edges give the action of elements of G . Take G to be free on generators a, b . Then get a hom. $\phi : G \rightarrow \mathbb{Z}/2\mathbb{Z}$ with $a \mapsto 0, b \mapsto 1$ and let $H = \ker \phi$. Then we have a graph with two vertices, and a takes each vertex to itself and b takes each vertex to the other. Thus elements of $H \iff$ paths from p to p , so H is the fund. group of this graph. Thus H is a free group on THREE gens which is bigger than 2 gens, the number of gens for G . The gens for H are

- (a) go along tree to start of edge
- (b) go along edge
- (c) go along tree back to point p

Thus H is free on $a, b^2, b^{-1}ab$.

45. **Free group on ∞ -gens subgroup of free group on 2 gens.** Let G be a free group on 2 gens a, b . Now make a graph that is an infinite row of discrete points. Let G act on this infinite row so that a takes a vertex and sends it to the vertex immediately on its right, and let b take a vertex to itself. Then H is the fundamental group of this graph, where H is generated by $b, a^{-1}ba, a^{-2}ba^2, \dots$
46. **Hom. from free group on 2 gens to S_3 .** Let G be a free group on 2 gens a, b . Define $\phi : G \rightarrow S_3$ so that $a \mapsto (12)$ and $b \mapsto (23)$. Put $H := \ker \phi$. S_3 is given by a, b modulo the relations $a^2 = 1 = b^2$ and $(ab)^3 = 1$. H is not a group generate dby $a^2, b^2, (ab)^3$, where the group generated by $a^2, b^2, (ab)^3$ is not normal. Now draw 6 points in a circle. Then a sends one vertex to an adjacent one (and vice-versa) and b does similarly but for alternating pairs of vertices compared to a . Then we get elements $a^2, abab^{-1}a^{-1}, \dots$, so in total, 7 elements generate H .
47. **Group is a category.** Any group is a category, where there is only 1 object (the group itself) and the morphisms are the elements of the group, and composition corresponds to the group product.
48. **Poset is a category.** Any a poset is a category, where the objects are the elements of the poset and the morphisms $\text{Mor}(A, B)$ has 1 element if $a \leq b$ and has 0 elements if otherwise.
49. **Topological space is a category (Grothendieck topologies).** Any topological space is a category, where the objects are the open sets and the morphism $\text{Mor}(A, B)$ has 1 element if $A \subset B$ and has 0 elements if otherwise. (this is also known as the Grothendieck topologies.)
50. **Nasty category example.** Take the category of commutative rings. Look at $\text{Mor}(\mathbb{Z}, Q)$. This is an epimorphism, but map on sets is not surjective. $\mathbb{Z} \xrightarrow{\text{id}} Q$ and $Q \xrightarrow{f} R$ and $Q \xrightarrow{g} R$ with $f = g$ on $\mathbb{Z}$ and $f = g$ on Q_i
51. **Category of categories.** The collection of all categories is a category, where the objects are categories and the morphisms are functors.
52. **Original example.** Let A be the category of topological spaces. Let B be the category of abelian groups. Then let H_i be the i th homology, and it is a functor.
53. **Forgetful functor example.** Let A be the category of groups and B be the category of sets. Then the functor $A \rightarrow B$ taking a group to its underlying set is a forgetful functor, as it “forgets” the group structure on the group.

54. **Forgetful functor example 2.** Let A be the category of rings and B be the category of abelian groups. Then the functor $A \rightarrow B$ taking a ring to its underlying abelian group is a forgetful functor, since it forgets the multiplication structure on the ring.
55. **Abelianization functor example.** Let A be the category of groups and B be the category of abelian groups. Then there is a functor $A \rightarrow B$ taking a group G to G/N , where N is the smallest normal subgroup so that G/N is abelian.
56. **Complex vector space example.** Take a functor from a group to the category of complex vector spaces. This corresponds to an action of G on complex vector spaces.
57. **Dual of a vector space (over $\mathbb{C}$).** Let F be a functor from category of all vector spaces to the category of all vector space duals. Then F is a contravariant functor.
58. **Homology functor.** Homology is a covariant functor, since if $X \rightarrow Y$, then $H_i(X) \rightarrow H_i(Y)$.
59. **Cohomology functor.** Cohomology is a contravariant functor, since if $X \rightarrow Y$, then $H_i(Y) \rightarrow H_i(X)$.
60. **Vector space to double dual space.** The diagram given by $V \xrightarrow{s} V$, $V \rightarrow V^{**}$, another $V \rightarrow V^{**}$, and $V^{**} \xrightarrow{s^{**}} V^{**}$ gives a commutative diagram.
61. **functor category.** If A, B are categories, then the functors from A to B form a category $\text{Fun}(A, B)$. 0-categories are just sets. Functors from a, b form a set. A category of categories forms a 2-category. Keep doing this, and we can see what n -categories are.
62. **Adjoint functor.** Let A be the category of groups and B be the category of sets. Then let $G : A \rightarrow B$ be a forgetful functor and $F : B \rightarrow A$ be the left adjoint of G . Then $F(S)$ is a free group generated by a set S . Then we have:

$$\begin{array}{ccc}
 H & & \text{underlying set of } H = G(H) \\
 \text{morphisms of groups} \downarrow & & \text{morphisms of sets} \downarrow \\
 \text{free group on } S = F(S) & & S
 \end{array}$$

63. **Forgetful adjoint functor.** Let A be the category of sets acted on by group H and let B be the category of sets. Then let F be a forgetful functor $F : A \rightarrow B$. Then F has a left adjoint $S \rightarrow S \times H$ and right adjoint $S \rightarrow \text{Fun}(H, S)$.

1. **Examples of rings without an identity 1.** (1): Ideals I of rings (say, $2\mathbb{Z}$). (2): example from analysis (all continuous functions on $\mathbb{R}$ with compact support).

2. **Examples of rings.** $\mathbb{Q}, \mathbb{R}, \mathbb{C}, \mathbb{Z}$, polynomial rings $R[x]$, matrix rings $M_n(R)$, $\mathbb{Z}[i]$.

3. **Analogues between Groups and Rings.**

Groups:

- (a) Act on sets (left, right, two-sided).
- (b) Symmetric group S_n (acts on n points).
- (c) Cayley's theorem that any group is the symmetric group of some object S with right G -action (acted on by G on the left).

Rings:

- (a) Act on modules ($R \times M \rightarrow M$, satisfying basic axioms) (with left, right, two-sided actions on modules).
- (b) R acts on R^n ($M_n(R)$ is the analogue of S_n).
- (c) Any ring is the set of endomorphisms of some abelian group A with some structure (where endomorphisms on A are of the form $f : A \rightarrow A$ where f is a homomorphism preserving $+$). So, $\text{End}(A)$ is a ring.

4. **Analogues between Sets and Free R -modules R^n .**

Sets:

- (a) disjoint union $S_1 \cup S_2$.
- (b) Cartesian product of sets.
- (c) $|A \cup B| = |A| + |B| - |A \cap B|$.
- (d) $|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$.

Free R -modules R^n .

- (a) some of free R -modules $R_1 \oplus R_2$.
- (b) tensor product of R_1, R_2 , which is $R_1 \otimes R_2$ (where if R_1 has a basis $e_1, \dots, e_m$ and R_2 has a basis $f_1, \dots, f_n$, then $R_1 \otimes R_2$ has a basis $e_i \otimes f_j$).
- (c) $\dim(V + W) = \dim V + \dim W - \dim(V \cap W)$.
- (d) FAILS (as shown in Axler problem from H110).

5. **Non-obvious examples of rings.**

NON-RING EXAMPLE Burnside ring of a group G . Look at sets acted on by G (up to isomorphism). Try to form into ring, by defining addition on 2 sets to be the disjoint union and product to be the Cartesian product of sets. Then let 0 be the empty set and 1 be the 1-element set acted on trivially. But this is not a ring as there is NO SUBTRACTION.

- (a) Grothendieck ring. Choose a collection of sets $S_1, \dots, S_n$ acted on transitively by G (up to isomorphism). Any finite set acted on by G is a union of some copies of $S_1, \dots, S_n$. We can write these as $a_1 S_1 + \dots + a_n S_n$ with $a_i \in \mathbb{Z}_{\geq 0}$. We can make this into a ring by letting $a_i \in \mathbb{Z}$.

6. **Burnside ring of S_3 .** Transitive actions of S_3 correspond to conjugacy classes of subgroups. We have subgroups of S_3 of order 1,2,3,6. For order 1, we have an action of 6 points, call it $\boxed{6}$. For order 2, we have 3 conjugate subgroups (for 3 points), call it $\boxed{3}$. For order 3, we have 1 subgroup (2 points), call it $\boxed{2}$. Same thing for order 6, 1 point, call it $\boxed{1}$. Then $\boxed{1}, \boxed{2}, \boxed{3}, \boxed{6}$ is a basis for the Burnside ring (look at diagram from notes).
7. **Group ring of a group (construction).** Let G be a group, and let its group ring be called $Z[G]$. Elements of $Z[G]$ are of the form $\sum n_i g_i$, with $n_i \in \mathbb{Z}$ and $g_i \in G$, and the g_i form a basis (over $\mathbb{Z}$) for $Z[G]$.
8. **Categorical construction of group ring of a group.** Suppose we have $Rings \rightarrow Groups$. Then let $A : Rings \rightarrow Groups$ be the forgetful functor taking a ring R to $R^\times$ (where $R^\times$ is the set of invertible elements (under $\times$) of R). Then we have a functor $B : Groups \rightarrow Rings$ so that B takes a group to its group ring construction (as the item above). Then B is left-adjoint to A . We can also say that ring homomorphisms take $Z[G]$ to R and group homomorphisms take G to $R^\times$.
9. **Examples of group rings.**
 - (a) Let $G = \mathbb{Z}$. Then $\mathbb{Z}[\mathbb{Z}]$ be the group ring of $\mathbb{Z}$, with basis x^n , for $n \in \mathbb{Z}$. Thus the group ring $\mathbb{Z}[\mathbb{Z}]$ is precisely $\mathbb{Z}[x, x^{-1}]$ of Laurent power series. Then $\mathbb{Z}[\mathbb{Z}_{\geq 0}] = \text{polynomial ring } \mathbb{Z}[x]$.
 - (b) $\mathbb{C}[\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}]$. Thus elements of this group ring are of the form $z_1 + z_2a + z_3b + z_4c$, where $1, a, b, c$ are elements of $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ (as a basis) and $z_i \in \mathbb{C}$. Note that $1 = (1, 0) + (0, 1)$. Put e as the first addend as f as the second addend. Then $ef = fe, e + f = 1, e^2 = e, f^2 = f$. So, e, f commute, are idempotents, and 1 is a sum of commuting idempotents. Thus we can write a ring R as a product of 2 other rings A, B by writing $R = A \times B$, for $A = eR$ and $B = fR$. Note also that $\mathbb{C}[\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}]$ has commuting idempotents other than 0, 1, by looking at $\frac{1+a+b+c}{4}, \frac{1+a-b-c}{4}, \frac{1+b-a-c}{4}, \frac{1+c-b-a}{4}$. Thus we see that assigning e_i to the respective elements gives that $e_1 + e_2 + e_3 + e_4 = 1$, $e_i e_j = e_j e_i$, and $e_i^2 = e_i$. Thus we can write $\mathbb{C}[\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}]$ as a direct sum of 4 copies of $\mathbb{C}$.
10. **COUNTEREXAMPLE to Co-product for Groups Injectivity Prop. (but for RINGS).** Suppose the Co-product for Groups Injectivity Prop. holds for rings. Then if R, S are rings and $R * S$ is their co-product, then for any ring X , we have the diagram with the maps $R \rightarrow X, S \rightarrow X, R \rightarrow R * S, S \rightarrow R * S, R * S \rightarrow X$ commutes, but it is not necessarily true that $R \rightarrow R * S$ is injective. Suppose $R \rightarrow R * S$ were injective. Then as in the proof of Co-product for Groups Injectivity Prop., thus the map $R \rightarrow S$ must be a ring homomorphism so that each $r \in R$ is sent to 0, but this is NOT a ring homomorphism. (can check this explicitly with $\mathbb{Z}/2\mathbb{Z} * \mathbb{Z}/3\mathbb{Z} = 0$).

11. **COUNTEREXAMPLE TO EVERYTHING IN REP THEORY.** Let $G = (\mathbb{Z}, +)$. Let $V = \mathbb{C} \oplus \mathbb{C}$. Let G act on V by $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$. (the problem is that this matrix has a multiple eigenvalue). Then we get an exact sequence $0 \rightarrow \mathbb{C} \rightarrow \mathbb{C} \oplus \mathbb{C} \rightarrow \mathbb{C} \rightarrow 0$, where the the in the first $\mathbb{C}$, we have that $\mathbb{Z}$ acts trivially and in the last $\mathbb{C}$, $\mathbb{Z}$ acts trivially. However, in $\mathbb{Z}$ acts nontrivially on $\mathbb{C} \oplus \mathbb{C}$.
12. **A variation.** Let $G = \mathbb{Z}/p\mathbb{Z}$ (with p prime) and let $V = (\mathbb{Z}/p\mathbb{Z})^2$. Let G act by the matrix $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$. Then we have again that the matrix to the p th power is the identity matrix, and we get a similar problem.
13. **Example of writing a decomposable rep / subrep.** Suppose V is a rep of G and V has a subrep W . Want to write $V = W \oplus U$ (where we need U to be acted on by G). Obvious attempt to do this is $U = W^\perp$. This works if the orthogonal complement is well-defined and invariant under G . So, the orthogonal complement $\perp$ needs to be a positive definite Hermitian form $(\cdot, \cdot)$ (with $(a, b) = \overline{(b, a)}$). V has $(\cdot, \cdot)$ but in general, not invariant. But we can make $(\cdot, \cdot)$ invariant if G is finite. Do this by $\sum_{g \in G} (ga, b)$, which is invariant positive definite form (not identically 0). Note that G must be finite (or compact) so the sum is defined.
14. **Decomposable rep example.** Let $G = \mathbb{Z}/2\mathbb{Z}$ with $G = \{1, \sigma\}$ and let it act on V . Then $V = V^+ \oplus V^-$, where on V^+ , σ acts as 1 and on V^- , σ acts as -1. We have that V^+ is a direct sum of 1-dim spaces and V^- is a direct sum of spaces. Write $v \in V$ as $v = \frac{v+\sigma(v)}{2} + \frac{v-\sigma(v)}{2}$, where the first term lies in V^+ and the second term lies in V^- . Thus G has 2 irreducible reps $(\mathbb{C}^1, \sigma = 1)$ and $(\mathbb{C}^1, \sigma = -1)$. We get a character table $\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$, with the first column label being 1 and the second column label being σ and the first row label being χ_0 and the second row label being χ_1 . The first row is the trivial rep and the second row is the nontrivial rep. The rows correspond to the irred reps. The columns correspond to the conjugacy classes of the group, which correspond to the elements (if G is abelian). The entries correspond to the traces of the elemnts of the rep (where a trace is a sum of eigenvalues).
15. **Representations of $\mathbb{Z}$.** Consider an action of $\mathbb{Z}$ on $\mathbb{C}^n$, which corresponds to a linear map from $\mathbb{C}^n \rightarrow \mathbb{C}^n$. We get Jordan normal form with Jordan blocks. Then $\mathbb{C}^n$ is the direct sum of subspaces corresponding to the Jordan blocks. The irred reps correspond to the Jordan blocks, which correspond to the nonzero eigenvalue λ plus an integer $n \geq 1$ (where n gives the size of the block). Also note that irreducibility corresponds with $n = 1$.
16. **Representations of $\mathbb{Z} \oplus \mathbb{Z}$.** The irreducibles are easy to classify. The irreducibles correspond to pairs (λ_1, λ_2) of nonzero complex numbers (eigenvalues of σ, τ) (with σ, τ generating $G = \mathbb{Z} \oplus \mathbb{Z}$).
17. **SU_{100} .** We can classify all reps of SU_{100} .

18. **Representations of S_3 .** Let $G = S_3$ (so $|G| = 6$) and G is nonabelian. We want to find at least some of its irreducible reps. There is a homomorphism $G \rightarrow \mathbb{Z}/2\mathbb{Z}$. So any rep of $\mathbb{Z}/2\mathbb{Z}$ gives a rep of G . So S_3 has 2 irred reps of dim 1 (1 trivial, 1 nontrivial). G acts on 3 points, a, b, c (permutation rep). This gives a linear rep $\mathbb{C}^3$ with basis a, b, c . Is this irreducible? (NO!!!). Notice that $a + b + c \in \mathbb{C}^3$ is fixed by G , so this gives a 1-dim subspace of $\mathbb{C}^3$ acted on by G . Its orthogonal complement is $\{xa + yb + zc \mid x + y + z = 0\}$. Notice that a, b, c is an orthonormal basis. Thus we get a 2-dim rep of G that is irred. Now we work out the character table of $G = S_3$. See character from class. The first row is the trivial rep, the second row is the sign rep, the third row is the 2-dim rep. Note that the first column gives 3 fixed points, the second column gives 1 fixed points, and the third column gives no fixed points. Note that we get the number of fixed points (as its own row) can be obtained by adding the rows (trivial rep) + (2-dim rep). If σ acts on S , what is its trace? If $|S| = 1$, then the matrix is (1) so the trace is 1. If $|S| > 1$, then $a \mapsto b \mapsto c \mapsto d \mapsto a$, so we get a matrix with 0's on the diagonal, 1's immediately above the diagonal and a 1 in the bottom left corner of the matrix, so the trace is 0. Thus, the trace of σ is the number of fixed points. Note that the columns of our character table are orthogonal so we have $\sum_{\chi} \chi(a)\overline{\chi(b)}$ for $a, b \in G$. The character is a row of the character table. The rows are not orthogonal (unless weighted by the number of elements in each conjugacy class). By weighting, we get $1 \cdot 1 \cdot 1 + 1 \cdot -1 \cdot 3 + 1 \cdot 1 \cdot 2 = 0$. So $\sum_{g \in G} \chi_1(g)\overline{\chi_2(g)} = 0$ if $\chi_1 \neq \chi_2$.
19. **Representations of $\mathbb{Z}/4\mathbb{Z}$.** $G = \mathbb{Z}/4\mathbb{Z}$ is generated by σ (with $\sigma^4 = 1$). The eigenvalues of σ can be $1, i, -1, -i$. Thus we get 4 irred reps. The character table is $\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & i & -1 & -i \\ 1 & -1 & 1 & -1 \\ 1 & -i & -1 & i \end{pmatrix}$, where the first column is the element $1 \in G$, the second column is $\sigma \in G$, the third column is $\sigma^2 \in G$, and the fourth column is $\sigma^3 \in G$. The rows are respectively $\chi_0, \dots, \chi_3$.
20. **Representations of $\mathbb{Z}/n\mathbb{Z}$.** Assume G is cyclic, so $G = \mathbb{Z}/n\mathbb{Z}$. Let G be generated by σ . G acts on $V = \mathbb{C}$. σ acts as multiplication by eigenvalue λ ($\lambda^n = 1$) so irred reps $\iff$ n th roots of 1. We get a character table with the first row being all 1's, the second row being $1, z, z^2, z^3, \dots$, the third row being $1, z^2, z^4, \dots$, and so on, where $z = e^{2\pi i/n}$.
21. **Vector space dual Canonical isomorphism ex.** Similar to finite-dim vector space of $\mathbb{C}$, we have that $V \simeq V^*$ (where V^* is the dual of V). However, this isomorphism is not canonical as it depends on the choice of basis. However, the isomorphism $V \simeq V^{**}$ is canonical. So $G \simeq G^{**}$ is canonical. if $g \in G$, $g^* \in G^*$, then $g^* \mapsto g^*(g)$ is a homomorphism from G^* to $\mathbb{C}$, so it is an element of G^{**} .
22. **Product of reps of a group G , where G is noncyclic.** Let G be a noncyclic group. Then $G = G_1 \times \dots \times G_n$, where G_i is cyclic. The representations of $G_1 \times G_2$ are homs $G_1 \times G_2 \rightarrow \mathbb{C}^*$, and we obtain $\text{Hom}(G_1 \rightarrow$

$\mathbb{C}^*) \times \text{Hom}(G_2 \rightarrow \mathbb{C}^*)$. Thus, we obtain the equation $|(G_1 \times G_2)^*| = |G_1 \times G_2|$.

23. **Fourier transform ex.** Let V be the vector space of all functions $G \rightarrow \mathbb{C}^*$. We get a Hermitian inner product $\langle a, b \rangle = \sum_{g \in G} a(g) \overline{b(g)}$. Characters form an orthogonal basis for V . Thus so if f is a function on G , we have $f = \frac{1}{|G|} \sum_{\chi} \chi(\chi, f)$ (where $\chi(\chi, f)$ are called Fourier coefficients), so f is a linear combination of characters. Thus, functions on G correspond to functions on G^* , and this correspondence gives Fourier transforms. Note that for Fourier series, we have that $f(x+1) = f(x)$, where f is a function on the circle group $\mathbb{R}/\mathbb{Z}$ (note that the circle group is compact). Note that for characters of the circle group, we can take $x \mapsto e^{2\pi i n x}$ for some $n \in \mathbb{Z}$. Thus we obtain $f(x) = \sum c_n e^{2\pi i n x}$.
24. **Dual of $\mathbb{Z}$.** The dual of $\mathbb{Z}$ is $\text{Hom}(\mathbb{Z}, \mathbb{C}^*) = \mathbb{C}^*$ (NOT $\mathbb{R}/\mathbb{Z}$). The dual of $\mathbb{Z}$ is defined using unitary characters. $\text{Hom}(\mathbb{Z}, S^1)$. Note that $S^1 \subset \mathbb{C}^*$, complex numbers z with $|z| = 1$.
25. **Dual of $\mathbb{Z}$ and dual of S^1 .** The dual of $\mathbb{Z}$ is S^1 . The dual of S^1 is $\mathbb{Z}$ (and we get Fourier series this way).
26. **$\mathbb{R}, \mathbb{Q}_p$ Fourier transform.** We have that $\mathbb{R}^* \simeq \mathbb{R}^*$ (by a Fourier transform) and $\mathbb{Q}_p^* \simeq \mathbb{Q}_p^*$.
27. **Things that go wrong about fields other than $\mathbb{C}$.** We have:
 - (a) over $\mathbb{R}$, $\mathbb{Z}/3\mathbb{Z}$ has irred reps of $\dim = 2 > 1$.
 - (b) over a field of characteristic 2, the group $\mathbb{Z}/2\mathbb{Z}$ has only 1 irred rep (the trivial rep), since any eigenvalue satisfies $\lambda^2 = 1$, and $\lambda^2 - 1 = (\lambda - 1)^2$ in characteristic 2, so $\lambda = 1$. Indecomposable reps of $\dim = 2$ give matrices like $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$.
28. **Application to Dirichlet's Theorem.** (see big long calculation in notebook).
29. **characters of $(\mathbb{Z}/a\mathbb{Z})^*$.** This is a finite abelian group of order $\phi(a)$. Recall that if $a = bc$, then $(\mathbb{Z}/a\mathbb{Z})^* \simeq (\mathbb{Z}/b\mathbb{Z})^* \times (\mathbb{Z}/c\mathbb{Z})^*$. If p is prime, then $(\mathbb{Z}/p\mathbb{Z})^*$ is cyclic. We have that $(\mathbb{Z}/p^n\mathbb{Z})^*$ is cyclic of order $(p-1) \cdot p^{n-1}$ if $p \neq 2$. Also, $(\mathbb{Z}/2^n\mathbb{Z})^* \simeq \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2^{n-2}\mathbb{Z}$.
30. **Character table for $(\mathbb{Z}/7\mathbb{Z})^*$.** A generator of the group is 3 (or 5). Thus, we see that for n -values 0,1,2,3,4,5, thus the 3^n -values reduce to 1,3,2,6,4,5, respectively. We get the following character table:

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & z & z^2 & -1 & z^4 & z^5 \\ 1 & z^2 & z^4 & 1 & z^2 & z^4 \\ 1 & -1 & 1 & -1 & 1 & -1 \\ 1 & z^4 & z^2 & 1 & z^4 & z^2 \\ 1 & z^5 & z^4 & 1 & z^3 & z^2 \end{pmatrix}$$

where the columns correspond to 1,3,2,6,4,5, respectively.

31. **Character table for $(\mathbb{Z}/8\mathbb{Z})^*$.** We get the character table for this (noncyclic) group:

$$\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}$$

where the columns correspond to 1, 3, 5, 7, respectively.

32. **Doing Application to Dirichlet's theorem but for primes congruent to 1 mod 4.** (see big long page in notebook).

33. **Character table for Quaternion group Q_8 .** The conjugacy classes are $1, -1, \pm i, \pm j, \pm k$. Also note that $Q_8/\pm 1 \simeq \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ (which are 4 1-dim characters). We get the character table:

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & -1 & -1 \\ 1 & 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 & 1 \\ 2 & -2 & 0 & 0 & 0 \end{pmatrix}$$

where the columns are for $1, -1, \pm i, \pm j, \pm k$, respectively. (We get the last row of values in the character table by orthogonality).

34. **Dihedral group D_8 (nonabelian) character table.** This group is D_8 , and there are 5 conjugacy classes. $|Z(D_8)| = 2$, so thus $D_8/Z(D_8) \simeq \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$. Thus the character table for D_8 is the same as the character table for Q_8 .

35. **Dihedral group D_{10} character table.** There are 4 conjugacy classes of D_{10} , where the conjugacy classes are the identity, 5 reflections, 2 rotations by $1/5$ of a revolution, and 2 rotations by $2/5$ of a revolution. Thus since we have 4 conjugacy classes, we get 4 characters. Thus we have $D_{10} \rightarrow \mathbb{Z}/2\mathbb{Z}$. The character table is:

$$\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & 1 \\ 2 & 0 & (2/5)\cos(2\pi) = z + z^{-1} & z^2 + z^{-2} \\ 2 & 0 & (2/5)\cos(4\pi) & (2/5)\cos(2\pi) \end{pmatrix}$$

where $z^5 = 1$. The first row corresponds to the trivial character. Also note that $10 = 1^2 + 1^2 + a^2 + b^2$, where a, b are the dimensions of the other characters.

36. **Character table for A_4 .** First find the conjugacy classes, which have order 1, 2, and 3. There are 1 elements in the group with order 1, 3 elements of order 2, and 4 elements of order 3, and another part of order 3 elements. Now find the 1-dim characters. Notice $A_4 = (\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}) \rtimes \mathbb{Z}/3\mathbb{Z}$, and so any character of $\mathbb{Z}/3\mathbb{Z}$ gives a character of A_4 . So the rows for the 1-dim characters are $1, 1, w, w^2$ and $1, 1, w^2, w$. The final character comes from orthogonality

(sum of squares of entries in a column must equal order of group), so we get $3, -1, 0, 0$, which is the tetrahedral character. We then get the character table (with the columns being order 1 conj class, order 2 conj class, an order 3 conj class, and another order 3 conj class):

$$\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & w & w^2 \\ 1 & 1 & w^2 & w \\ 3 & -1 & 0 & 0 \end{pmatrix}$$

37. **Character table for S_4 .** The conjugacy classes are given by $\text{id}, (12), (12)(34), (123), (1234)$. For 1-dim characters, we have that $1 \rightarrow A_4 \rightarrow S_4 \rightarrow \mathbb{Z}/2\mathbb{Z} \rightarrow 1$, so we get a sign character $1, -1, 1, 1, -1$. We get the third row from permutation representation. The fourth row comes from $\chi_2 \otimes \chi_1$. The fifth row comes from orthogonality. So we get the character table (with the conjugacy classes in the order as given above):

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & 1 & -1 \\ 4 & 2 & 0 & 1 & 0 \\ 3 & 1 & -1 & 0 & -1 \\ 2 & 0 & 2 & -1 & 0 \end{pmatrix}$$

note $\chi_4((12)(34)) = \dim \chi_4 = \chi_4(1)$. Through $\chi_4((12)(34))$, we get sum of 2 roots of unity (which are eigenvalues) and there are only 2 of them; this is only possible if 2 eigenvalues are both 1. So, the element is trivial. In 2-dim rep, elements of form $(12)(34), (13)(24), (14)(23)$ all act trivially. So they form a normal subgroup $= \ker(S_4) \rightarrow M_2(\mathbb{C})$. (NOTE THAT WE CAN FIND THE NORMAL SUBGROUPS BY LOOKING AT CHARACTER TABLE AND SEEING IF THERE ANY REPEATS WITHIN A CERTAIN ROW).

38. **Character table of binary tetrahedral group (order 24).** We write the binary tetrahedral group as $2T$. The order of $2T$ is 24. The conjugacy classes are $1, -1, i, a, b, c, d$. The cardinalities of the conj classes are $1, 1, 6, 4, 4, 4, 4$, respectively. Then let ξ be a third root of unity. Then we get the character table:

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & \xi & \xi^2 & \xi^2 & \xi \\ 1 & 1 & 1 & \xi^2 & \xi & \xi & \xi^2 \\ 3 & 3 & -1 & 0 & 0 & 0 & 0 \\ 2 & -2 & 0 & \xi & \xi^2 & -\xi^2 & -\xi \\ 2 & -2 & 0 & \xi^2 & \xi & -\xi & -\xi^2 \\ 2 & -2 & 0 & 1 & 1 & -1 & -1 \end{pmatrix}$$

39. **Character table for A_5 .** The group A_5 is the group of rotations of icosahedron. The conjugacy classes are given by $1, (12)(34), (123), (12345), (12354)$, and the conj classes have cardinalities $1, 15, 20, 12, 12$, respectively. (note that these cardinalities are different if we had taken S_5 instead of A_5). The first row

(character) χ_0 is the trivial character. The second row χ_1 is given by (permutation rep) - (trivial rep). The third row is given by $\chi_1 \otimes \chi_1 = V \otimes V$, and we get that $\mathbb{Z}/2\mathbb{Z}$ corresponds to the eigenspaces of $\mathbb{Z}/2\mathbb{Z}$. Eigenspace of $+1$ has dimension $10 = \frac{4 \times 5}{2}$ (related to symmetric square S^2V) and the eigenspace of -1 has dimension $6 = \frac{4 \times 3}{2}$ (related to alternating square). Look at the character of the symmetric square. Let $g \in G$. Eigenvalues are $\lambda_1, \dots, \lambda_n$ (with eigenvectors $v_1, \dots, v_n$). Then symmetric square has a basis $v_i \otimes v_j + v_j \otimes v_i$ ($i < j$). Then eigenvalues on symmetric square are $\lambda_i \lambda_j$ ($i \leq j$). Then $\sum_{i \leq j} (\lambda_i \lambda_j) = (1/2)((\sum \lambda_i)^2 + (\sum \lambda_i^2))$ and $v_i \otimes v_i$, which equals $(1/2)(\text{trace}(g)^2 + \text{trace}(g^2))$. So character of symmetric square of a 4-dim rep is $10, 2, 1, 0, 0$. Subtract known reps. $(\chi_{S^2V}, 1) = 1$ and $(\chi_{S^2V}, \chi_1) = 1$. So $\chi_{S^2V} = \chi_0 + \chi_1 + (\text{something})$. The character of (something) is given by $(10, 2, 1, 0, 0) - (1, 1, 1, 1, 1) - (4, 0, 1, -1, -1) = (5, 1, -1, 0, 0)$. Thus $(\chi_0, \chi_2) = 1$. Look now at the alternating square character. We get that $\chi_{\lambda^2(4)}(g) = (1/2)(\chi(g)^2 - \chi(g^2))$. There is a $-$ sign for λ^2 (the alternating square) and a $+$ sign for S^2 (the symmetric square). The row for $\chi_{\lambda^2(4)}$ is $6, -2, 0, 1, 1$. Notice $(\chi, \chi) = 2$ ($\chi = \oplus 2$ irred reps, where we have 2 missing irreducibles). $(\chi, \chi_0) = (\chi_1, \chi_1) = (\chi_1, \chi_2) = 0$. Since $1^2 + 4^2 + 5^2 + d_1^2 + d_2^2 = 60$, only solution is $d_1 = d_2 = 3$. Thus, $\chi_3 = (3, -1, 0, \frac{1+\sqrt{5}}{2}, \frac{1-\sqrt{5}}{2})$. Also, $\chi_4 = (3, -1, 0, \frac{1-\sqrt{5}}{2}, \frac{1+\sqrt{5}}{2})$. We obtain the first 3 columns of the last two rows of the character table by orthogonality. Notice in our character table we get a 2×2 block of irrational numbers.

40. Example where # (irred reps) NOT EQUAL TO # (conj classes).

Let $G = \mathbb{Z}/3\mathbb{Z}$, and the field be $\mathbb{R}$. Then there are 2 irred reps and 3 conjugacy classes. Another example is taking $G = \mathbb{Z}/2\mathbb{Z}$, the field be $\mathbb{Z}/2\mathbb{Z}$, then we have 1 irred rep and 2 conj classes.

41. application of algebraic integer. We have $\dim V \mid |G|$, since $\frac{|G|}{\chi(1)} =$

$\sum_g \frac{\overline{\chi(g)}\chi(g)|C|}{\chi(1)}$, where each addend in the sum is an algebraic integer so the sum is an algebraic integer. Since $|G|/\chi(1)$ is rational and also an algebraic integer, then it is an integer. However, we have that $\dim V$ need not necessarily divide $|G|$ for real reps. COUNTEREXAMPLE: take $\mathbb{Z}/3\mathbb{Z}$ (order 3) has irred 2-dim reps over $\mathbb{R}$, but 2 does not divide 3.

42. Find all simple groups of order ≤ 100 . We can assume G isn't cyclic.

By Burnside, $|G|$ divisible by at least 3 primes, and they can't all be odd, as $3 \cdot 5 \cdot 7 > 100$. So must be divisible by 2. So $|G|$ divisible by $2^2 = 4$ as any group of order $(2 \times \text{odd})$ has normal subgroup of index 2. The only possible cases we get are order $60 = 2^3 \cdot 3 \cdot 5$ and order $84 = 2^2 \cdot 3 \cdot 7$ groups. First look at order 84 case. Sylow theorems give that number of sylow 7-subgroups is 1, so it is normal, so G not simple. Thus we move on to the case where $|G| = 60$. (Want to show $G \simeq A_5$). Sylow thms gives G has 6 5-sylow subgroups, so 24 elements elements of order 5, and G has 10 sylow 3-subgroups and so, 20 elements of order 4. We also get that there are 16 elements of order 1, 2, or 4. It is enough to show G acts nontrivially on some set of size ≤ 5 , since then $G \leq S_n$ ($n \leq 5$). Any two 2-sylow subgroups are disjoint. If $z \in P$, $z \in Q$ (P, Q are sylow 2-subgroups), then centralizer of z has order > 4 and is divisible by

4. So number of conjugates of z is ≤ 5 , so G acts on a set of size ≤ 5 . So, can assume 2-sylow subgroups are disjoint (except for at the identity element). Thus we have at most 5 sylow 2-subgroups. Each has 3 nonidentity elements. G has ≤ 15 elements of order 1,2,4. Thus ≤ 5 sylow 2-subgroups, as these are disjoint. So G acts on set of size ≤ 5 , so $G = A_5$.

RING THEORY.

1. **Example of Euclidean rings.** These include $\mathbb{Z}$ and $k[x]$, where k is a field (and the Euclidean valuation for a polynomial in $k[x]$ is its degree).
2. **Example of a PID that IS NOT a Euclidean ring.** Take $R = \mathbb{Z}[\frac{\sqrt{-19}+1}{2}]$. (proof that it is PID is too advanced). We now show why it is not a Euclidean ring. Suppose for contradiction R is a Euclidean domain. Pick $a \in R$ so that $a \neq 0$ (and a not a unit) and $|a|$ small as possible subject to this constraint. Then in $R/(a)$, every element represented by 0 or a unit, since if $x \in R/(a)$, write $x = aq + r$ ($|r| < |a|$), so $r = 0$ or a unit, by choice of a . If $R = \mathbb{Z}[\frac{\sqrt{-19}+1}{2}]$, then its only units are ± 1 , so the number of elements that are 0 or unit is three. If $a \in R$, $a \neq 0$ and a not a unit, then $|R/(a)| \geq 4$, which is a contradiction.
3. **Example of a non-principal ideal.** Look at $\mathbb{R}[[x, x^{1/2}, x^{1/4}, x^{1/8}, \dots]]$. Put $x = (x^{1/2})^2$, $x^{1/2} = (x^{1/4})^2$, and so on. So, x not divisible by an irreducible element, $x \neq 0$, and x not a unit. The ideal generated by $x, x^{1/2}, x^{1/4}, \dots$ is not a principal ideal.
4. **Example of a UFD.** Take $\mathbb{Z}[i] = \{mi + n \mid m, n \in \mathbb{Z}\}$. We show that $\mathbb{Z}[i]$ is a UFD. (It is Euclidean). Define $|m + ni| := m^2 + n^2$ (square of norm), so it is an integer. Now imagine $\mathbb{Z}[i]$ as an integer lattice in $\mathbb{C}$. Then about each point, draw a circle around it with radius 1. These circles give open discs of radius 1 centered at each lattice point. Key observation: these open discs cover all of $\mathbb{C}$ (we use this to show that $\mathbb{Z}[i]$ is Euclidean). Suppose we are given $a, b \in \mathbb{Z}[i]$, $b \neq 0$. Want to find $a = qb + r$ ($|r| < |b|$). Thus $a/b = q + r/b$. Choose q so that $|q - a/b| < 1$, because $\mathbb{C}$ is covered by these open discs, so thus $|r|/|b| < 1$, so $|r| < |b|$. So, $\mathbb{Z}[i]$ is a UFD. If $n = a^2 + b^2 \iff n = (a + ib)(a - ib)$.
5. **Example of Fermat's theorem of sum of squares.** Fermat's theorem says that if p is prime and $p \equiv 1 \pmod{4}$, then $p = a^2 + b^2$ for some integers a, b . (examples are $5 = 2^2 + 1^2, 13 = 3^2 + 2^2, 17 = 4^2 + 1^2$). Now we prove the theorem. Suppose $p \equiv 1 \pmod{4}$. Thus -1 is a square modulo p . So $a^2 + 1 = np$ for some a, n , so p not irreducible in $\mathbb{Z}[i]$. $np = (a + i)(a - i)$. So p has a nontrivial factorization $p = (x + iy)(x - iy) \implies p = x^2 + y^2$.
6. $\mathbb{Z}[\sqrt{-2}]$. Applying the same logic (viewing $\mathbb{Z}[\sqrt{-2}]$ as a lattice in $\mathbb{C}$ and drawing circles of radius 1 about each lattice point) shows that the discs still cover $\mathbb{C}$. So we conclude $\mathbb{Z}[\sqrt{-2}]$ is a UFD. Note $(x + y\sqrt{-2})(x - y\sqrt{-2}) = x^2 + 2y^2$, so we get theorems about primes of the form $x^2 + 2y^2$.
7. $\mathbb{Z}[\sqrt{-3}]$. Applying the same covering circles of the lattice logic shows that the circles intersect at the point $\frac{\sqrt{-3}+1}{2}$. Thus $\mathbb{C}$ is covered by closed discs (radius

1), NOT BY OPEN DISCS of radius 1. This ring is not Euclidean, does not have unique factorization. We see that $(1 + \sqrt{-3})(1 - \sqrt{-3}) = 2 \cdot 2$, where each factor on each side of the equation is an irreducible element, so we get non-unique factorization. Note that $|ab| = |a| \cdot |b|$, so if $|1 + \sqrt{-3}| = 4 = |2|$, then $\mathbb{Z}[\sqrt{-3}]$ has no elements x with $|x| = 2$. Also $1 + \sqrt{-3}$ is not a unit and neither is 2.

8. **Examples of nonprincipal ideals.** A typical principal ideal looks like $a\mathbb{Z}[\sqrt{-3}]$, whose points form a rectangular lattice in $\mathbb{C}$. Take $(2, 1 + \sqrt{-3})$, which is a nonprincipal ideal. Its lattice is diamond-shaped, so not principal, since by Eisenstein, $\mathbb{Z}[\frac{1+\sqrt{-3}}{2}]$ is a UFD. $\mathbb{C}$ is covered by open discs. $\mathbb{Z}[\sqrt{-3}] \subset \mathbb{Z}[\frac{1+\sqrt{-3}}{2}]$.

9. **Example of a Non-UFD.** Take R to be the ring of all holomorphic functions. THIS IS NOT A UFD! Take $\sin(z)$ in R . The irreducible elements of R are those functions with one zero. By Weierstrass, any holomorphic function is an infinite product of irreducibles (unique up to order and units).

10. **Picturing commutative rings.**

(a) Method 1. Draw a point for each element of the ring, so if $R = \mathbb{R}$, then we draw a line, for $\mathbb{Z}[i]$, we draw a square lattice, etc., and we can see that the latter is a UFD.

(b) Method 2. Draw a point for each basis element of the ring. (for instance, for $R = \mathbb{C}[x]$, draw a point for each of $1, x, x^2, x^3, \dots$) In $\mathbb{C}[x]$, every ideal is generated by 1 element. In $\mathbb{C}[x, y]$, is every ideal is generated by 2 elements? NO!!!. Based on this method, the picture for $\mathbb{C}[x, y]$ is a square lattice in quadrant 1 of $\mathbb{R}^2$, with the horizontal axis having the points $1, x, x^2, \dots$ and the vertical axis having the points $1, y, y^2, \dots$. We can take an “inner diagonal” of the lattice and see that all elements not inside the inner diagonal are generated by the elements in the inner diagonal. Thus we see that $\mathbb{C}[x, y]/(x^3, x^2y, xy^2, y^3)$ is a ring of dim 6 over $\mathbb{C}$.

(c) Method 3. (works for all rings) Draw a point for each maximal or prime ideal.

11. **Image of ring Hom maximal.** Suppose we have a homomorphism $g : R \rightarrow S$ of rings. We would like to get a homomorphism of maximal spectrums. Suppose M is a maximal ideal of S . Then the image of g should be ideal of points of elements $r \in R$ so that $f(r) \in M$. PROBLEM: (NOT MAXIMAL, i.e. the ideal of elements $r \in R$ so that $f(r) \in M$ is not maximal). For example, take $f : \mathbb{Z} \rightarrow \mathbb{Q}$ to be a ring homomorphism. Then let $M = (0)$ (the zero ideal) of S . Then M is maximal in S , but $f^{-1}(0) = 0$, which is not maximal in $\mathbb{Z}$.

12. **Prime spectrum and prime ideal.** We have that (prime spectrum of R) $\iff$ (prime ideals). (topology has a basis $D(f) = \{\text{prime ideals } P \text{ with } f \notin P\}$). We get a contravariant functor (commutative rings) $\rightarrow$ (topological spaces). For instance, if R is a field, then $\text{Spec}(R) = \text{point}$.

13. **Finding $\text{Spec}(\mathbb{C}[x])$.** Take $\mathbb{C}[x]$. The prime ideals of it are $(x - \alpha)$ ($\alpha \in \mathbb{C}$) and (0) . Thus $\text{Spec}(\mathbb{C}[x]) = \mathbb{C} \cup \{0\}$, which is not closed. The topology is

really weird (not Hausdorff, non-closed point)! A closed set of polynomial $f =$ set of zeroes of f which is a finite set of points.

14. **Prime ideals of $\mathbb{Z}$.** Take $R = \mathbb{Z}$. Its prime ideals are $(0), (2), (3), (5), (7), \dots$. Closed sets are finite subsets of $(2), (3), (5), \dots$ + whole space. We had a homomorphism $\mathbb{Z} \rightarrow \mathbb{Q}$. We get a map $\text{Spec}(\mathbb{Q}) \rightarrow \text{Spec}(\mathbb{Z})$, with $(0) \mapsto (0)$. Visualizing this gives $(2), (3), (5), \dots$ are just points whereas (0) is a 1-dimensional point (a line on which $(2), (3), \dots$ are sitting).
15. **Tensor product examples.** Take $R = \mathbb{Z}$. Easy properties are $(M_1 \oplus M_2) \otimes N \cong (M_1 \otimes N) \oplus (M_2 \otimes N)$ and $R \otimes_R R \cong R$. So this gives $M \otimes N$ for M, N free abelian groups. (Any finitely generated abelian group is a direct sum of $\mathbb{Z}$, $\mathbb{Z}/n\mathbb{Z}$)
16. **Tensor product calculation example.** $\mathbb{Z}/2\mathbb{Z} \otimes \mathbb{Z}/3\mathbb{Z} = 0$, and $\mathbb{Z}/12\mathbb{Z} \otimes \mathbb{Z}/10\mathbb{Z} = \mathbb{Z}/2\mathbb{Z}$.
17. **Tensor product preserves exact sequences (COUNTEREXAMPLE).** The Tensor product does not preserve exact sequences. Consider the counterexample: $0 \rightarrow \mathbb{Z} \xrightarrow{\times 2 (\text{injective})} \mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z} \rightarrow 0$ and suppose we take the tensor product with $\mathbb{Z}/2\mathbb{Z}$ (with the written sequence) and we get $0 \rightarrow \mathbb{Z}/2\mathbb{Z} \xrightarrow{\times 2} \mathbb{Z}/2\mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z} \rightarrow 0$, but now the $\times 2$ map is not injective.
18. **Eratosthenes method for finding primes in $\mathbb{Z}$ and in $F_2[x]$.** List all positive integers $2, 3, 4, 5, \dots$. Circle 2, then cross out all multiples of it. Go to the next noncrossed out number, circle it, then cross out all multiples of it, and so on. We apply the same method in $F_2[x]$, where F_2 is the field with 2 elements, $\mathbb{Z}/2\mathbb{Z}$. Write all polynomials in $F_2[x]$, so we have $x, x+1, x^2, x^2+1, x^2+x, x^2+x+1, x^3, x^3+1, \dots$. Note that multiples of $f \in F_2[x]$ have $\sum \text{coeffs} = 1$ and with constant term 0. So we can look at when $\sum \text{coeffs} = 1$ and constant term = 1 to find all the primes, so we get $x^3+x+1, x^3+x^2+1, \dots$ (see notes 10/21/25 for more).
19. **Polynomials in $\mathbb{Z}/8\mathbb{Z}[x]$.** We had a theorem that if f has degree n , then it has at most n roots. But if we take $f = x^2 - 1 \in \mathbb{Z}/8\mathbb{Z}[x]$, then we see that f has $4 > 2$ roots, namely, $1, 3, 5, 7$, but the problem is that $\mathbb{Z}/8\mathbb{Z}$ has zero-divisors.
20. **Polynomials in ring of quaternions, $\mathbb{H}$.** We had a theorem that if f has degree n , then it has at most n roots. But if we take $f = x^2 + 1 \in \mathbb{H}[x]$, then it has roots $\pm i, \pm j, \pm k$, and so it has uncountably many roots since then any $ai + bj + ck$ (with $a^2 + b^2 + c^2 = 1$) is a root, as from class. But the problem here is that $\mathbb{H}$ is not commutative.
21. **Application of the result that degree n polys have at most n roots.** Let F be a field, G a finite subgroup of F (under multiplication). Then G is cyclic. To prove this, see that $x^n - 1$ has $\leq n$ roots, so G has $\leq n$ elements of order dividing n . G is abelian, so thus G is cyclic, and this follows from fundamental theorem of finite abelian groups. THIS FAILS FOR QUATERNIONS!!!!

22. **Calculating content of polynomial.** Take $6x^3 + 10x + 8$. Its content is 2, since $2 = \gcd(6, 8, 10)$. $c(f = 2x^3 + 10/3) = 2/3$, since then $f/c(f) = 3x^3 + 5$, where 3, 5 are coprime coeffs.
23. **Prime dividing content of a product of polys.** Suppose p is a prime of $\mathbb{Z}$, and $p \mid fg$, with $f, g \in \mathbb{Z}[x]$. Then $p \mid c(fg) = c(f)c(g)$, so $p \mid c(f)$ or $p \mid c(g)$, so $p \mid f$ or $p \mid g$, so thus p is prime in $\mathbb{Z}[x]$.
24. **Content of an irred poly in $\mathbb{Z}[x]$.** Suppose $f \in \mathbb{Z}[x]$ is irred with $c(f) = 1$. Suppose $f \mid gh$ in $\mathbb{Z}[x]$. Then $f \mid gh$ in $\mathbb{Q}[x]$, so $f \mid g$ in $\mathbb{Q}[x]$ or $f \mid h$ in $\mathbb{Q}[x]$, since $\mathbb{Q}[x]$ is a UFD. Suppose we say that $f \mid g$ in $\mathbb{Q}[x]$. Then $fj = g$ in $\mathbb{Q}[x]$. $c(f) = 1$ and $c(g) \in \mathbb{Z}$, so $c(j) \in \mathbb{Z}$, so $j \in \mathbb{Z}[x]$, so $f \mid g$ in $\mathbb{Z}[x]$.
25. **Kronecker algorithm to factor in $\mathbb{Z}[x]$.** Take $f(x) \in \mathbb{Z}[x]$ with $\deg(f) = n$. Then any poly of $\deg \leq n$ is determined by its values at $n + 1$ points, say $0, 1, \dots, n$. Suppose $f = g$. Look at $f(0), \dots, f(n)$, and say that $f(i) = g(i)h(i)$ for all i , $0 \leq i \leq n$ and suppose $f(i) \neq 0$ for all i , $0 \leq i \leq n$. Then we see that $g(i) \mid f(i)$, so only finite number of possibilities for $g(0), \dots, g(n)$, which determines g . Thus g can be found in finite time.
26. **Methods to Test to see if a Polynomial is irreducible.**
- (a) can use a probabilistic test and say that for degree (greater than 0) polynomials, they are usually irred.
 - (b) if $f \in \mathbb{Z}[x]$ has leading coeff. 1 and is irred. in $F_p[x]$, then f is irred. (as $\deg f \pmod{p} = \deg f$) For instance, $31x^3 + 5x + 1$ is irred as $x^3 + x + 1$ is irred modulo 2. Also $2x^2 + x$ is irred modulo 2, but is reducible $\mathbb{Z}[x]$.
27. **Applying Eisenstein's criterion.** We can apply Eisenstein's criterion to cyclotomic polys. Take $x^p - 1 = (x^{p-1} + x^{p-2} + \dots + 1)(x - 1)$, and we want to show that $x^{p-1} + x^{p-2} + \dots + 1$ is irred. Trick: change $x \mapsto x + 1$. Thus we see that $\frac{(x+1)^p - 1}{x} = x^p + (p \text{ choose } 1)x^{p-1} + (p \text{ choose } 2)x^{p-2} + \dots + (p \text{ choose } 1)$, and all of the coeffs (except the leading coeff) are divisible by p and the last term is divisible by p but not divisible by p^2 .
28. **Unexpected factorizations.** Can use Aurifellian method to factorize $2^{58} + 1 = 4 \times (2^{14})^4 + 1 = 4a^4 + b^4 = (2a^2 + 2ab + b^2)(2a^2 - 2ab + b^2)$.
29. **Another way to test irred of a poly in $\mathbb{Z}[x]$.** If $f = x^3 + ax^2 + bx + c \in \mathbb{Z}[x]$, then it is irred if no factors of c are roots of f . As an example, we see that $x^3 - 3x + 1$ is irred (as ± 1 are not roots).
30. **Method for factorization of polys over $F_p[x]$.** See last part of notes from 10/21/25.
31. **Example of a NON-NOETHERIAN RING.** Let k be a field. Then take $R = k[x_1, x_2, x_3, \dots]$. Then take the ideal $I = (x_1, x_2, x_3, \dots)$. This ideal I is NOT FINITELY GENERATED, so thus R is NOT Noetherian.
32. **Example of the TFAE Ring Theorem.** Take $R = k[x_1, x_2, \dots]$.

- (a) R is clearly not Noetherian as seen from class, as we have that $I = (x_1, x_2, \dots)$ is not finitely generated.
- (b) take $\{(x_1), (x_1, x_2), (x_1, x_2, x_3), \dots\}$, which is a nonempty set that has no maximal element.
- (c) We have an infinite increasing chain, namely $(x) \subsetneq (x_1, x_2) \subsetneq (x_1, x_2, x_3) \subsetneq \dots$
33. **Infinite strictly decreasing chain.** Take $R = \mathbb{Z}$. Then look at the infinitely strictly decreasing chain given by $(2) \supset (4) \supset (8) \supset \dots$, and this set of ideals has no minimal element.
34. **Application: Finite Generation of Invariant Rings.** Suppose G is a group acting on $\mathbb{C}^n$. Look at polys. $\mathbb{C}[x_1, \dots, x_n]$ (also acted on by G). Look at $\mathbb{C}[x_1, \dots, x_n]^G$, which is the ring of polynomials fixed by action of G , and $\mathbb{C}[x_1, \dots, x_n]^G$ is called the ring of invariants. Problem: is this algebra finitely generated? Answer is often yes (Hilbert), but sometimes no (Nagata). See last 2 or 3 pages of notebook class notes from 10/23/25.
35. **Example of invariant ring action by group.** Take $G = S_3$. Then G acts on $\mathbb{C}^3$. Then $G = S_3$ acts on polynomials on $\mathbb{C}^3 \cong \mathbb{C}[x_1, x_2, x_3]$. The ring of invariants is $\mathbb{C}[x_1 + x_2 + x_3, x_1x_2 + x_2x_3 + x_3x_1, x_1x_2x_3]$. G acts on graded poly. ring $\mathbb{C}[x_1, \dots, x_n]$ (graded by degree). See last 2 or 3 pages of notebook class notes from 10/23/25.
36. **Example of invariants.** Let $G = S_n$ act on $\mathbb{C}^n$. Let $P(v) = \text{polys} = \mathbb{C}[x_1, \dots, x_n]$. Then $P(v)^G = \text{symmetric polys}$. For instance, for $n = 3$, these symmetric polynomials are $x_1 + x_2 + x_3, x_1x_2 + x_2x_3 + x_3x_1, x_1^3x_2 + x_1^3x_3 + x_2^3x_1 + \dots, \dots$
37. **Sum of k th powers example.** Suppose α, β, γ are roots of $z^3 + z + 1 = 0$. Want to find $\alpha^k + \beta^k + \gamma^k$. Note that $p_0 = 3, p_1 = 0$, where p_1 is the coeff of z^2 . Then $p_2 + p_0 = 1, p_3 + p_1 + p_0 = 0, p_4 + p_2 + p_0 = 0, p_5 + p_3 + p_2 = 0$, so $p_0 = 3, p_1 = 0, p_2 = -2, p_3 = -3, p_4 = -1, p_5 = 5$.
38. **Applicatoin of p_k to Adams Operations ψ^k .** Here, we do the case where $k = 2$. Take a vector space V . Split $V \otimes V$ as $S^2(V) \oplus \lambda^2(V)$. $\psi^2(V) = S^2(V) \ominus \lambda^2(V)$. Elements of $S^2(V)$ look like $a \otimes b + b \otimes a$, and elements of $\lambda^2(V)$ look like $a \otimes b - b \otimes a$. The dimension of $S^2(V)$ is $n(n+1)/2$ and the dimension of $\lambda^2(V)$ is $n(n-1)/2$. The eigenvalues of $S^2(V)$ are $\lambda_i\lambda_j$ with $i \leq j$ and the eigenvalues of $\lambda^2(V)$ are $\lambda_i\lambda_j$ with $i < j$. If g is an automorphism of V with eigenvalues λ_i , then g has eigenvalues λ_i^2 on $\psi^2(V)$.
39. **Discriminant calculation ($n = 3$).** We are in the $n = 3$ case. Suppose we have a polynomial in e_1, e_2, e_3 , namely the polynomial $x^3 - e_1x^2 + e_2x - e_3$. Then $\Delta^2 = 18e_1e_2e_3 - 4e_1^3e_3 + e_1^2e_2^2 - 4e_2^3 - 27e_3^2$.
40. **Discriminant calculation ($n = 2$).** If we have the polynomial $x^2 - bx + c$, then $\Delta^2 = b^2 - 4c$.

41. **Resultant calculation example.** Take the polynomial $f(x) = x^3 + bx + c$. Then $f' = 3x^2 + b$. We get the following Sylvester matrix:

$$\begin{pmatrix} 1 & 0 & b & c & 0 \\ 0 & 1 & 0 & b & c \\ 3 & 0 & b & 0 & 0 \\ 0 & 3 & 0 & b & 0 \\ 0 & 0 & 3 & 0 & b \end{pmatrix}$$

where the first two rows correspond to f and the last three rows correspond to f' . Taking the determinant of this Sylvester matrix yields $\pm(4b^3 + 27c^2)$. To pin down constant factor, look at explicit f , say $f(x) = x^3 - x = (x + 1)(x)(x - 1)$. Thus the discriminant of this explicit choice for $f(x)$ is $-4b^3 - 27c^2$ (APPROXIMATELY).

42. **Algebraic Geometry Discriminant example.** Take an elliptic curve of the form $y^2 = x^3 + bx + c$. When is it nonsingular? It is nonsingular whenever $\Delta(x^3 + bx + c) = 0$, i.e. $-4b^3 - 27c^2 \neq 0$.
43. **Number Theory Discriminant example.** The discriminant of a cubic field of $\mathbb{Q}[\alpha]$, where $\alpha^3 + b\alpha + c = 0$ is $\Delta = (\text{square}) \times (\text{discriminant of } x^3 + bx + c = 0)$.
44. **Discriminant example.** The discriminant of $x^3 + x + 1$ is APPROXIMATELY $-4 \cdot 1^2 - 27^2 = -31$.
45. **Example of computing quotients with formal power series.** Let $R = k[x]$ and $I = (x)$, where k is a field. Then $R/I = \{a_0\}$, with $a_i \in k$. Then $R/I^2 = \{a_0 + a_1x\}$, with $x^2 = 0$. Then $R/I^3 = \{a_0 + a_1x + a_2x^2\}$, with $x^3 = 0$, and so on. If $a_0 + a_1x + a_2x^2$ is a formal power series, then we get $a_0 \in R/I$, $a_0 + a_1x \in R/I^2$, $a_0 + a_1x + a_2x^2 \in R/I^3$, and so on. Also note that we get a chain of maps $\cdots \rightarrow R/I^3 \rightarrow R/I^2 \rightarrow R/I$. Note that this is a compatible set of elements of R/I^k for all $k \geq 1$. We can do the same for any R, I and the result is called the completion of R at ideal I .
46. **Example of completion of ring at ideal.** Let $R = k[x_1, x_2, x_3]$, where k is a field, and let $I = (x_1, x_2, x_3)$. Then its completion is the power series $R[[x_1, x_2, x_3]]$.
47. **Completion of the ring of integers with respect to ideal (p) .** Let $R = \mathbb{Z}$ and let $I = (p)$, where p is a prime. What is the completion? (the completion is the p -adic integers) We have maps $\mathbb{Z}/p\mathbb{Z} \leftarrow \mathbb{Z}/p^2\mathbb{Z} \leftarrow \mathbb{Z}/p^3\mathbb{Z} \leftarrow \cdots$. We have that in base p , $a_0 \in \mathbb{Z}/p\mathbb{Z}$, $a_1a_0 \in \mathbb{Z}/p^2\mathbb{Z}$, $a_2a_1a_0 \in \mathbb{Z}/p^3\mathbb{Z}$, and so on. The number in base p with infinite number of digits before the “decimal point” gives the p -adic integers. Why do we take p prime? Suppose instead we had the 10-adic integers (where 10 isn’t prime). Then we have maps $\mathbb{Z}/10\mathbb{Z} \leftarrow \mathbb{Z}/100\mathbb{Z} \leftarrow \cdots$, and we notice that $\mathbb{Z}/10\mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/5\mathbb{Z}$ is a map, and $\mathbb{Z}/100\mathbb{Z} \rightarrow \mathbb{Z}/2^2\mathbb{Z} \times \mathbb{Z}/5^2\mathbb{Z}$ is a map, and so on. Thus we get that the 10-adic integers is really $(2\text{-adic integers}) \times (5\text{-adic integers})$.

48. **A $\mathbb{C}$ -example of formal power series.** Take the set of polynomials $\mathbb{C}[x_1, \dots, x_n]$, which is a subset of (ring of holomorphic functions on $\mathbb{C}^n$), which is a subset of (ring of formal power series), which thus gives an algebraic analog of holomorphic functions.
49. **Units of ring of formal power series.** Take 1-variable example. Take $R = k[[x]]$, with k a field. What are its units? Note $(1+x)^{-1} = 1 - x + x^2 - x^3 + \dots$. Then see that $(1 + a_1x + a_2x^2 + \dots)^{-1} = (1+b)^{-1}$ (where b is $a_1x + a_2x^2 + \dots$), which is then $= 1 - b + b^2 - b^3 + \dots$, and the b -term gives $x()$, the b^2 -term gives $x^2()$, the b^3 -term gives $x^3()$, and so on. Note that the alternating sum of b converges to the inverse of $1 + a_1x + a_2x^2 + \dots$. In general, $a_0 + a_1x + a_2x^2 + \dots$ has inverse IFF a_0 has an inverse ($a_0 \neq 0$).
50. **Example of formal power series.** $1 + x/2 + x^2/2^2 + x^3/2^3 + \dots \in \mathbb{Q}[[x]]$.
51. **Application of Weierstrass prep. theorem to proving $k[[x, y]]$ is a UFD.** (the proof for polynomials fails, as content is not defined for power series) Here, we omit routine details and just prove key step: if f is irreducible, then f is prime. Suppose f is irred in $k[[x, y]]$ and $f \mid gh$. WTS $f \mid g$ or $f \mid h$. First, we can assume f, g, h are Weierstrass polys in x (multiplication by units or by y^* is harmless), so $fr = gh$ (by assumption that $f \mid gh$) for some $r \in k[[x]]$. Put $r = pu$, where p is Weierstrass poly and u is a unit. Then $(fp) \cdot u = gh$, where fp is a Weierstrass ply and u is a unit, and gh is a Weierstrass poly decomposition of gh . So by uniqueness of Weierstrass decomposition of gh , must have $fp = gh$, $u = 1$. So $r = p$ is poly in x (not just power series in x). So f divides gh in $k[[y]][x]$, which is polys in x with coeffs in $k[[y]]$. Note that f is irred in $k[[y]][x]$. Note that $k[[y]][x]$ is a UFD, since poly ring over UFD $k[[y]]$. Thus irreds are prime, so f divides g or h in $k[[y]][x]$, so in $k[[x, y]]$. So, irreds are prime.
52. **Counterexample to claim that if f, g polys and $f \mid g$ in $k[[x]]$, then $f \mid g$ in $k[x]$.** For instance take $f = 1+x$ and $g = 1$. Then $1/f = 1/(1+x) \in k[[x]]$, but $1/(1+x) \notin k[x]$, so even though f, g are polys and $f \mid g$ in $k[[x]]$, we have that f does not divide g in $k[x]$.
53. **Counterexample to claim that if f is a poly in $k[x, y]$ and is irred, it is irred in $k[[x, y]]$.** For instance take $y^2 - x^3 - x^2$. Then graphing this gives a (see picture in notes) that has only 1 component, so it is irreducible in this sense. However, zooming in near the origin shows that the graph looks like an X and so we have a factorization $(y + x\sqrt{1+x})(y - x\sqrt{1+x})$.

1. **Examples of fields.** Examples of fields are $\mathbb{R}, \mathbb{C}, \mathbb{Q}, \dots$. From number theory, $\mathbb{Q}[i]$ is an algebraic number field. From algebraic geometry, for an elliptic curve $y^2 = x^3 + ax + b$, can consider the coordinate ring $\mathbb{C}[x, y]/(y^2 - x^3 - ax - b)$, which is an integral domain, and we get a field of fractions out of this.

2. **Field of fractions examples.** An example of a field of fractions is that the field of fractions of $\mathbb{Z}$ is $\mathbb{Q}$. We have a map $\mathbb{Z} \rightarrow \mathbb{Q}$, where $\mathbb{Z}$ is an integral domain and $\mathbb{Q}$ is a field, and similarly another map is (holomorphic functions) $\rightarrow$ (meromorphic functions on $\mathbb{C}$).
3. **Examples of residue fields (quotient fields).** Examples of quotient fields include $R/(\text{maximal ideal})$, $\mathbb{Z}/p\mathbb{Z}$, $k[x]/(f)$ (f irreducible), $\mathbb{Q}[i] = \mathbb{Q}[x]/(x^2 + 1)$. Now look at $k[x]/(f)$, but what happens if f is not irreducible? Then write $f = gh$, with g, h irreducible ($g \neq h$). Then $k[x]/(f(x)) = k[x]/g \times k[x]/h$ (Chinese remainder theorem). Note that since f is not irreducible, thus $k[x]/(f)$ has zero divisors. As an example, notice that $\mathbb{Z}/6\mathbb{Z} = \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$. Also, if f is not irreducible and $f = g^2$, then $k[x]/(f)$ has nilpotent elements ($f \neq 0, f^2 = 0$).
4. **Example of field extension.** $\mathbb{R} \subset \mathbb{C}$ is a field extension and $[\mathbb{C} : \mathbb{R}] = 2$. Also, $\mathbb{Q} \subset \mathbb{R}$ is a field extension with $[\mathbb{R} : \mathbb{Q}] = \infty$.
5. **Algebraic / Transcendental example.** Note that $\sqrt{2}$ is algebraic over $\mathbb{Z}$. x is algebraic over K in field $K(x)$ (where $K(x)$ is the field of fractions of $K[x]$). Note that π is transcendental over $\mathbb{Q}$ (HARD to show, omitted in class).
6. **Finding a polynomial with specified root.** It is REALLY HARD to find a polynomial with $\sqrt{2} + \sqrt{3} + \sqrt{5} + \sqrt{7}$ as a root (skipped in class, because it is too hard).
7. **Silly example of e, π transcendental.** e, π are known to be transcendental. Open problems are it is unknown if $e\pi$ and $e + \pi$ are both transcendental. We know that one of $e\pi, e + \pi$ is not algebraic, since e, π are roots of $x^2 - (e + \pi)x + e\pi = 0$, so if $e\pi, e + \pi$ are both algebraic, then so are e, π , which we know they can't be.
8. **Splitting field examples.** Let the field be $\mathbb{Q}$. Let $f(x) = x^3 - 2$. Then a splitting field is $\mathbb{Q}[i] = \mathbb{Q}[x]/(x^2 + 1)$. Now let $f(x) = x^3 - 2$, which has roots $2^{1/3}, w \cdot 2^{1/3}, w^2 \cdot 2^{1/3}$, where w is a third root of unity. Then we could try $\mathbb{Q}[x]/(x^3 - 2) \cong \mathbb{Q}[2^{1/3}] \subset \mathbb{R}$. Note that $\mathbb{Q}[2^{1/3}]$ contains only 1 of the 3 roots. In this quotient, f factors as $(x - 2^{1/3})(x^2 + 2^{1/3}x - 4^{1/3})$, and the polynomial (as the second factor) is irreducible in $\mathbb{Q}[2^{1/3}][x]$. Now get a splitting field $\mathbb{Q} \subset \mathbb{Q}[2^{1/3}] \subset \frac{\mathbb{Q}[2^{1/3}][x]}{(x^2 + 2^{1/3}x - 4^{1/3})}$, and now we see that $x^3 - 2$ factors in this quotient.
9. **Another splitting field example.** Take $f(x) = x^4 + 1$, which is irreducible over $\mathbb{Q}$. Then try $\mathbb{Q} \subset \mathbb{Q}[x]/(x^4 + 1) \cong \mathbb{Q}(\frac{1+i}{\sqrt{2}})$. Other roots of $x^4 + 1$ are w^3, w^5, w^7 , and these other roots are already in $\mathbb{Q}(w)$, where w is a 4th root of unity. Note that $\mathbb{Q}[x]/(x^4 + 1)$ is already a splitting field.
10. **Finite field of order 4.** Take $F_2[x]/(x^2 + x + 1)$, which has 4 elements and $x^2 + x + 1$ is the only irreducible polynomial of degree 2. The elements of $F_2[x]/(x^2 + x + 1)$ are $1, 0, x, x + 1$. WARNING! The additive group is NOT $\mathbb{Z}/p^n\mathbb{Z}$, but is $(\mathbb{Z}/p\mathbb{Z})^n$.

11. **Proof of existence of finite fields.** Put L to be the splitting field of $f = x^{p^n} - x$. Then $f'(x) = -1$, so $\gcd(f, f') = 1$, so no multiple roots, so all roots are distinct, so thus it has p^n roots. AMAZING FACT is that these p^n roots form a field. Closed under multiplication: $x^{p^n} - x = x(x^{p^n} - 1)$, which has a root at 0 and the roots of the second factor are $p^n - 1$ roots of 1, and these $p^n - 1$ roots form a group under multiplication. Closed under addition: $x^{p^n} = F \circ F \circ \dots \circ F(x)$, where $F(x) = x^p$ is the Frobenius map. Key point is that $F(x + y) = F(x) + F(y)$, since $(x + y)^p = x^p + y^p$ if $p = 0$ in the field. So $(x + y)^{p^n} - (x + y) = (x^{p^n} - x)(y^{p^n} - y)$. So if x, y roots, so is $x + y$. So we have a field of order p^n : all roots of $x^{p^n} - x$ in splitting field. This also shows that the field of order p^n is a splitting field, so it is UNIQUE up to isomorphism.
12. **Find an explicit field of order 2^4 .** Look at splitting field of $x^{16} - x$. How do we construct this? Need to find an irreducible factor of $x^{16} - x$. Note that $x^{16} - x = x(x - 1)(x^2 + x + 1)(x^4 + x^3 + 1)(x^4 + x + 1)(x^4 + x^3 + x^2 + x + 1)$. Its splitting field is $F_2[x]/(x^4 + x + 1)$, and we choose the ideal $(x^4 + x + 1)$. But the choice of factor (in the quotient) is NOT UNIQUE, as we could have instead picked $F_2[x]/(x^4 + x^3 + 1)$ or $F_2[x]/(x^4 + x^3 + x^2 + x + 1)$. These are all valid options. How to write down factorization of $x^{16} - x$? If f is any irreducible polynomial of degree 4 over F_2 , then $f \mid (x^{16} - x)$. Note that $F_2[x]/(f)$ has order 16, so it is isomorphic to splitting field. If $\deg f$ divides 4, then f divides $x^{16} - x$. All irreducible polynomials of degree dividing 4 (monic).
13. **Algebraic closure of a field K .** Take the splitting field of the set of all polynomials over K , and we need to define this set. The splitting field of a set of polynomials is obvious. Splitting field exists and is unique (up to isomorphism). Algebraic closure of K : algebraically closed means all polynomials split $K_0 \subset K_1 \subset K_2 \subset \dots \subset K_\infty$ (K_{n+1} splitting field of all polynomials in $K_n[x]$). Note that K_1 is the splitting field of all polynomials. In fact $K_1 = K_2 = \dots = K_\infty$ is algebraically closed. So it is an algebraic closure of K_0 .
14. **Non-normal extension example.** $\mathbb{Q} \subset \mathbb{Q}(2^{1/3})$ is a degree 3 extension. Adjoin a root of $x^3 - 2$, the other roots of $x^3 - 2$ are complex and thus not in this extension.
15. **degree 2 extensions are normal.** Any degree 2 extension is normal, i.e. when $K \subset L$ and $[L : K] = 2$. Suppose $f \in K[x]$ is irred, and has a root $\alpha \in L$. If $f(x) = ax^2 + bx + c$, then $\sum(\text{roots}) = -b/a$, so the other roots is $-b/a - \alpha \in L$.
16. **$x^4 + 1$ roots example.** What about $\mathbb{Q}(\alpha)$, where α is a root of $x^4 + 1 = 0$? Let α be an 8th root of 1 and α a root of $x^4 + 1 = 0$. Then the other roots of $x^4 + 1 = 0$ are $\alpha^3, \alpha^5, \alpha^7$, so it $x^4 + 1$ factors as $(x - \alpha)(x - \alpha^3)(x - \alpha^5)(x - \alpha^7)$. Is $\mathbb{Q}(\alpha)$ normal? What about other irreducible polynomials with root in $\mathbb{Q}(\alpha)$? The answer becomes clearer with more propositions and theorems, see other document.
17. **Example of normal extension of normal extension is not normal.** The extension $\mathbb{Q} \subset \mathbb{Q}(\sqrt{2})$ is normal, since it is of degree 2. The extension

$\mathbb{Q}(\sqrt{2}) \subset \mathbb{Q}(2^{1/4})$ is normal. But the extension $\mathbb{Q} \subset \mathbb{Q}(2^{1/4})$ is not normal, because $\mathbb{Q}(2^{1/4})$ has roots $\pm 2^{1/4}$ and $\pm 2^{1/4}i$, but these second two complex roots are not in $\mathbb{Q}(2^{1/4}) \subset \mathbb{R}$.

18. **Example of normal subgroup of a normal subgroup is not normal.** $\mathbb{Z}/2\mathbb{Z} \subset \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ is a normal subgroup, as it has index 2. $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z} \subset D_8$ is also a normal subgroup. But $\mathbb{Z}/2\mathbb{Z} \subset D_8$ is not a normal subgroup. Note that D_8 is the Galois group of $x^4 - 2$.
19. **Example of a non-separable extension.** Take $x^p - a$ in $K[x]$, where $\text{char} K = p > 0$ and suppose $x^p - a$ is irred, so it has a root $a^{1/p}$ in L . Note that $x^p - a = (x - a^{1/p})^p$, so all roots of $x^p - a$ are the same.
20. **Finite field extension are separable.** If $K \subset L$ and K, L are finite fields, then the extension is separable. Proof: if $f \in K[x]$ irred of degree n , then $f \mid (x^{p^n} - x)$, which has derivative -1 , so it is separable.
21. **$\mathbb{R} \subset \mathbb{C}$ Galois correspondence.** Take the field extension $\mathbb{R} \subset \mathbb{C}$. Let $G = \text{Gal}(\mathbb{C}/\mathbb{R}) = \{\text{id}, \text{conj}\}$. G has 2 subgroups, the trivial subgroup and G itself. The Galois correspondence is order-reversing, i.e. (BIGGER subgroups H) $\iff$ (SMALLER extensions L).
22. **Example of Galois correspondence.** Take $x^3 - 2$. Note that $\mathbb{Q} \subset (\text{splitting field of } x^3 - 2)$. Note $\mathbb{Q} \subset \mathbb{Q}(2^{1/3}) \subset \mathbb{Q}(2^{1/3}, w)$, where the first extension is degree 3 and the second extension is degree 2, thus the complete extension is degree 6 (where w satisfies $w^2 + w + 1 = 0$), as it is a splitting field of a separable polynomial. $G = S_3$ is the permutations of the 3 roots $2^{1/3}, 2^{1/3}w, 2^{1/3}w^2$, and we get the symmetry group of an equilateral triangle in $\mathbb{C}$ by looking at roots of unity. Now we look at the correspondence between subgroups and subfields. Note that $(12), (13), (23)$ each are index 2 subgroups of G (which are NOT NORMAL) and we get a group generated by $1, (123), (132)$ as an index 3 subgroup of G , which is NORMAL. Then these have index 3 and index 2 (respectively) in S_3 (see diagram in notes). For the subfields, we have $\mathbb{Q}(2^{1/3}, w)$ at the top with subfields $\mathbb{Q}(2^{1/3}), \mathbb{Q}(2^{1/3}w), \mathbb{Q}(2^{1/3}w^2)$, which are degree two with respect to $\mathbb{Q}(2^{1/3}, w)$ and these are NOT NORMAL over $\mathbb{Q}$. We also have $\mathbb{Q}(w)$ a degree 3 extension with respect to $\mathbb{Q}(2^{1/3}, w)$, which is a normal over $\mathbb{Q}$. Thus these have degree 3 and 2 over $\mathbb{Q}$ (see diagram in notes).
23. **Galois group of a finite field.** Suppose we have $F_2 \subset F_{16}$. What is the Galois group of F_{16} ? The Galois group of F_{16} is generated by Frobenius F with $F(x) = x^2$. Note that $F^4(x) = (((x^2)^2)^2)^2 = x^{16}$. Thus $F^4 = 1$. So the Galois group of F_{16} is cyclic of order 4, so it is isomorphic to $\mathbb{Z}/4\mathbb{Z}$. Also we get the following Galois correspondence. For subgroups, note that $\mathbb{Z}/4\mathbb{Z} \supset \mathbb{Z}/2\mathbb{Z} \supset 1$ and correspondingly, $F_2 \supset F_4 \supset F_{16}$.
24. **Galois group of $x^7 - 1$ over $\mathbb{Q}$.** Note $x^7 - 1 = (x - 1)(x^6 + x^5 + \cdots + x + 1)$, where the second factor on the RHS is irreducible (see a previous example on cyclotomic polynomials). If w is a root, so is $w^2, w^3, \dots, w^6$. So $\mathbb{Q} \subset \mathbb{Q}(w)$ is a degree 6 Galois extension. What is the Galois group? It is order

6. Note that possible homomorphisms are given by $w \mapsto w, w \mapsto w^2, \dots, w \mapsto w^6$ (where $(w^i)^j = w^{ij}$), and thus we conclude that the Galois group is $(\mathbb{Z}/7\mathbb{Z})^*$. Recall that $(\mathbb{Z}/7\mathbb{Z})^*$ is cyclic and generated by 3 (or 5) and has order 6, so it is isomorphic to $\mathbb{Z}/6\mathbb{Z}$, and it thus has 4 subgroups. Elements of $\mathbb{Z}/6\mathbb{Z}$ are 0, 1, 2, 3, 4, 5, and the corresponding elements of $(\mathbb{Z}/7\mathbb{Z})^*$ are 1, 3, 2, 6, 4, 5. Subgroups of $\mathbb{Z}/6\mathbb{Z}$ are 0, (0, 3), (0, 2, 4), (0, 1, 2, 3, 4, 5). Subgroups of $(\mathbb{Z}/7\mathbb{Z})^*$ are 1, (1, 6), (1, 2, 4), (1, 2, 3, 4, 5, 6). Thus we see that out of 1, we get subgroups (1, 6) and (1, 2, 4), with the former having index 2 (and this subgroup (1, 6) corresponds to complex conjugation). Then in (1, 2, 3, 4, 5, 6), the subgroup (1, 6) has index 3. For the corresponding subfields, we have that $\mathbb{Q}(w)$ has a degree 2 subfield $\mathbb{Q}(w) \cap \mathbb{R}$, which is degree 3 over $\mathbb{Q}$, and this subfield is fixed by conjugation, as it is an intersection with $\mathbb{R}$. We want to find elements fixed by a group H . Take average to get $\sum_{g \in H} g(*)$. Find elements fixed by (1, 2, 4). Set $a = w + w^2 + w^4$, where w is a “random element”. What is a ? Note that $a^2 = w^2 + w^4 + w^8 + w + 2w^3 + 2w^5 + 2w^6 = -2 - a$ (note: $1 + w + \dots + w^6 = 0$). Thus $a^2 + a + 2 = 0$, so by quadratic formula, $a = \frac{-1 \pm \sqrt{-7}}{2}$, so thus the mystery field (so far missing in the correspondence) is $\mathbb{Q}\sqrt{-7}$ (note: the sign inside the square root depends on $7 \equiv 3 \pmod{4}$, and since 7 is a prime).

25. **Construct heptadecagon with ruler & compass.** This is a 17-sided polygon. We do it with Galois theory. Find the roots of $x^{17} - 1 = 0$, using $+, -, \times, \div, \sqrt{}$, and note that $\sqrt{}$ yields a degree 2 extension. We have (for cyclotomic polynomials) $x^{17} - 1 = (x - 1)(1 + x + x^2 + \dots + x^{16})$, and the second factor in the RHS has degree $16 = 2^4$. The Galois group is $(\mathbb{Z}/17\mathbb{Z})^* \cong \mathbb{Z}/16\mathbb{Z}$. Note that elements of $\mathbb{Z}/16\mathbb{Z}$ are 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, and the corresponding elements of $(\mathbb{Z}/17\mathbb{Z})^*$ are 1, 3, 9, 10, 13, 5, 15, 11, 16, 14, 8, 7, 4, 12, 2, 6. We want to find fields $\mathbb{Q} \subset K \subset L \subset M \subset \mathbb{Q}(w)$, where each of the consecutive extensions is degree 2 and $w^{17} = 1$. In $\mathbb{Z}/16\mathbb{Z}$, say $(0, 4, 8, 12) \iff L$. In $(\mathbb{Z}/17\mathbb{Z})^*$, $(1, 9, 13, 15, 16, 8, 4, 2) \iff K$. Want to describe K, L, M explicitly. K is generated by $a = w^1 + w^9 + w^{13} + w^{15} + w^{16}$. We find that $a^2 + a - 4 = 0$, so $a = \frac{-1 \pm \sqrt{17}}{2}$. So, $K = \mathbb{Q}(\sqrt{17})$. What is L ? L is generated by $b = w^1 + w^{13} + w^{16} + w^4$, which corresponds to the subgroup (1, 13, 16, 4). We want $b^2 + *b + * = 0$, where $* \in K$. Put $c = w^9 + w^{15} + w^8 + w^2$, where we have that 9, 15, 8, 2 is a coset of 1, 3, 16, 4 in the subgroup (1, 9, ...). Note that $b + c$ and bc are both fixed by group of order 8, so thus $b + c, bc \in K$. So, b, c are roots of a quadratic equation in $K[x]$. We get $bc = -1, b + c = w + w^9 + \dots = \frac{-1 + \sqrt{17}}{2}$. We deal with M and $\mathbb{Q}(w)$ similarly. This procedure also works for primes $2^n + 1$ (not just $p = 17$), and these are called Fermat primes (3, 5, 17, 257, 65537, ...).
26. **Splitting field of $x^4 - 2$ over $\mathbb{Q}$.** Note that the splitting field is $\mathbb{Q}(2^{1/4}, i)$, which is degree 8 over $\mathbb{Q}$. Notice that the Galois group is D_8 , the symmetry group of a square. (See diagram in class notes for Galois subgroup $\iff$ subfield correspondence)
27. **Find Galois field extension given a Galois group.** Suppose we are given a finite group G . Can we find a Galois extension $K \subset M$ with Galois group G ? (YES) Take $\mathbb{Q}(t_1, \dots, t_n)$ field of all rational functions of $t_1, \dots, t_n$ acted on by S_n (permutations of t_i). We take $G \subset S_n$ for some n , by Cayley's theorem.

Then see $\mathbb{Q}(t_1, \dots, t_n)^G \subset \mathbb{Q}(t_1, \dots, t_n)$ is a Galois extension with Galois group G . We now ask a slightly harder question: Can we take K, M finite extensions of $\mathbb{Q}$? (YES) We just have to find $\mathbb{Q} \subset M$ with $\text{Gal}(M/\mathbb{Q}) = S_n$. Then take $Q \subset M^G \subset M$ (with $M^G \subset M$ Galois, group G). It is enough to take p prime. We just do the case with $p = 5$. The splitting field of $x^5 - 4x + 2$ has Galois group S_5 , order 120. Notice:

- (a) $x^5 - 4x + 2$ is irreducible by Eisenstein, where G is the Galois group of the splitting field. See that $5 \mid [\text{splitting field} : \mathbb{Q}]$, so $5 \mid |G|$, and $G \subset S_5$, so G contains a 5-cycle.
 - (b) The Galois group contains complex conjugates. The polynomial $x^5 - 4x + 2$ has 3 real, 2 complex roots. So, complex conjugation is a 2-cycle. So $G \subset S_5$ contains a 5-cycle and a 2-cycle, so $G = S_5$. So, $G = S_5$, so we found a Galois extension of $\mathbb{Q}$ with Galois group S_5 .
28. **Splitting field of $x^5 - 4x + 2$ continued.** This polynomial has Galois group S_5 over $\mathbb{Q}$. Note that $\mathbb{Q} \subset L$ (where this is the corresponding extension) has Galois group S_5 and so $[L : \mathbb{Q}] = 120$. Take now H to be a subgroup of S_5 . Then look at $\mathbb{Q} \subset L^H \subset L$. $L^H \subset L$ is a normal Galois extension with Galois group H but $\mathbb{Q} \subset L^H$ is usually not a normal extension. (The only normal subgroups of S_5 are 1, A_5 , and itself). Given G , can we find Galois extension $\mathbb{Q} \subset L$ with Galois group G ? (related to the inverse Galois problem).
29. **$G = \mathbb{Z}/5\mathbb{Z}$, find Galois extension.** Let $G = \mathbb{Z}/5\mathbb{Z}$. Can we find a Galois extension of $\mathbb{Q}$ with Galois group G ?
- (a) METHOD 1: for $x^{11} - 1 = (x - 1)(x^{10} + \dots + x + 1)$, the Galois group of its splitting field is $(\mathbb{Z}/11\mathbb{Z})^* \cong \mathbb{Z}/10\mathbb{Z}$ and we get an onto map $\mathbb{Z}/10\mathbb{Z} \rightarrow \mathbb{Z}/5\mathbb{Z}$. We get an exact sequence $0 \rightarrow \mathbb{Z}/2\mathbb{Z} \rightarrow \mathbb{Z}/10\mathbb{Z} \rightarrow \mathbb{Z}/5\mathbb{Z} \rightarrow 0$. For $\mathbb{Q} \subset \mathbb{Q}(w)^H$, its degree is 5 (and hence get Galois group $G/H = \mathbb{Z}/5\mathbb{Z}$) and for $\mathbb{Q}(w)^H \subset \mathbb{Q}(w)$, its degree is 2, where $w^{11} = 1$ and H is a subgroup of order 2. We have $\text{Gal}(\mathbb{Q}(w)^H/\mathbb{Q}) = \mathbb{Z}/5\mathbb{Z} = \text{Gal}(\mathbb{Q}(w)/\mathbb{Q})$, and H is the elements fixing all elements of $\mathbb{Q}(w)^H$. H is the subgroup with $w \mapsto w^{-1}$. This is complex conjugation. So $\mathbb{Q}(w)^H$ is a real subfield of $\mathbb{Q}(w)$, which is $\mathbb{R} \cap \mathbb{Q}(w)$, which is generated by (say) $w + w^{-1} = e^{2\pi i/11} + e^{-2\pi i/11} = 2\cos(2\pi/11)$, and notice that $w + w^{-1}$ is fixed by H .
 - (b) METHOD 2: use $x^{25} - 1 = (x^5 - 1)(x^{20} + \dots + 1)$, and the right factor on the RHS is irreducible, by Eisenstein. The Galois group is $(\mathbb{Z}/25\mathbb{Z})^*$, which is cyclic of order 20, so it has a quotient isomorphic to $\mathbb{Z}/5\mathbb{Z}$.
30. $(\mathbb{Z}/5\mathbb{Z})^2$. Using method 1 (as in the previous example) gives $\mathbb{Z}/5\mathbb{Z}$ for any prime $\equiv 1 \pmod{5}$. Method 2 gives 1 extension $\mathbb{Z}/5\mathbb{Z}$. Problem is that we need to find lots of primes $\equiv 1 \pmod{5}$. Dirichlet's theorem, we recall, says that there are infinitely many primes $\equiv a \pmod{b}$ if $\gcd(a, b) = 1$. From this, we conclude that we have extensions $\mathbb{Q} \subset L$ with Galois group $(\mathbb{Z}/p\mathbb{Z})^n$ for $n > 0$.
31. **Galois group of degree 3 polynomial.** Let $f(x) = x^3 + ax^2 + bx + c$ and assume f is irreducible. Splitting field has degree divisible by 3, and also

divides 6. So it has degree 3 or 6. If it is 3, then the Galois group is $\mathbb{Z}/3\mathbb{Z}$. If it is 6, then the Galois group is S_3 . Problem: does the splitting field have degree 3 or 6? To do this, look at $\Delta := (\alpha - \beta)(\beta - \gamma)(\gamma - \alpha)$, where α, β, γ are the roots of f . Fixed by order 3 subgroup $\alpha \mapsto \beta \mapsto \gamma \mapsto \alpha$. So Galois group is $A_3 = \mathbb{Z}/3\mathbb{Z}$ iff Δ is in the splitting field K . If the Galois group is not A_3 , then $(\alpha\beta)$ takes Δ to $-\Delta$, so Δ is not in base field ($\text{char} K \neq 2$). Check if Δ^2 is a square (we know that Δ^2 is a square is equivalent to saying Δ is in the base field) (where Δ is the discriminant of f). So Galois group is $\mathbb{Z}/3\mathbb{Z}$ or S_3 ; it is $\mathbb{Z}/3\mathbb{Z}$ if Δ^2 is a square and it is S_3 if Δ^2 is not a square.

32. $x^3 + bx + c$ **depressed cubic Galois group.** Let $f(x) = x^3 + bx + c$, with $\Delta^2 = 4b^3 - 27c^2$. Taking $x^3 - 3x - 1$ gives $\Delta^2 = 81 = 9^2$. So Galois group is $\mathbb{Z}/3\mathbb{Z}$. Take now $x^3 + 3x - 1$. Then $\Delta^2 = -135 \neq n^2$, so here, the Galois group is S_3 .

33. **Show that $\mathbb{C}$ is algebraically closed.**

- (a) proof 1: use Liouville's theorem from complex analysis.
- (b) take $f(z)$ polynomial, see diagram from class regarding winding number.
- (c) Almost algebraic proof: (the non-algebraic facts used for this proof are that (1) any poly of odd degree has a root (IVT) and (2) any complex number has a square root). Fact (1) implies $\mathbb{R}$ has no extensions of odd degree (since if $\mathbb{R}$ did have an extension of odd degree, then we get an irreducible poly of odd degree, which can't happen). Let K be a finite Galois extension of $\mathbb{R}$, containing $\mathbb{C}$. Then we know G has no subgroups of odd index. So $|G|$ is a power of 2. (since any sylow 2-subgroup has odd index, so sylow 2-subgroup is G). Put H to be the subgroup of index 2 corresponding to $\mathbb{R} \subset \mathbb{C}$. Then H has no subgroups of index 2, as $\mathbb{C}$ has no extensions of degree 2. So $|H| = 2^n$ and H has no subgroups of index 2, so $|H| = 1$. So $|G| = 2$ and Galois extension is $\mathbb{C}$. Thus $\mathbb{C}$ is algebraically closed.

34. **Given Galois group, what is L ?** Problem: suppose Galois group G of $K \subset L$ is known. What can we say about L ? Take $G = \mathbb{Z}/2\mathbb{Z}$. Then $[L : K] = 2$, so L is generated by root of $x^2 + bx + c = 0$. Complete the square to get $(x + b/2)^2 + c - (b/2)^2 = 0$, so we have the form $x^2 = a$, so $L = K[\sqrt{a}]$, if $\text{char} K \neq 2$. We had a problem as we had to divide by 2 (had to divide b by 2) in the completing the square step. What if $\text{char} K = 2$? Take $x^2 + bx + c = 0$. If $b = 0$, the inseparable, so NOT a Galois extension. If $b \neq 0$, change x to bx , then $x^2 + x + * = 0$, so $L = K(\alpha)$, where α is a root of $\alpha^2 - \alpha = \text{constant}$ (this equation is called the Artin-Schrier poly). Now suppose $G = \mathbb{Z}/3\mathbb{Z}$. Let $K \subset L$. If $\alpha \in L$, then α is a root of a cubic poly. Magic word is EIGENVECTOR. Look at Galois group generated by σ , (with $\sigma^3 = 1$). Think of σ has a linear map of 3-dim vector space L . Try to pick an eigenvector of σ . Eigenvalues are $1, w, w^2$ (with $w^3 = 1$), and we want $w \in K$. Assume $x^3 - 1$ splits in K . For eigenvalue 1: (fixed by σ) we have $v + \sigma(v) + \sigma^2(v)$ (each addend is an element of K) with $v \in L$. For eigenvalue w : fixed by $w^{-1}\sigma$ and have $v + w^{-1}\sigma + w^{-2}\sigma^2(v)$. For eigenvalue w^2 : we have

$v + w^{-2}\sigma + w^{-1}\sigma^2(v)$. Is every element of L a sum of eigenvectors? (add them up) Sum is $3v + 0\sigma v + 0\sigma^2 v = 3v$ (where $1 + w + w^2 = 0$). So v is a sum of eigenvectors if $3 \neq 0$, i.e. if $\text{char} K \neq 3$. So we can find $\alpha \in L$ so that α is an eigenvector with eigenvalue $\neq 1$, say $\sigma\alpha = w\alpha$. $\sigma\alpha^3 = w^3\alpha = \alpha$, so σ fixes α^3 , so $\alpha^3 \in K$, so α is a root of $x^3 - a = 0$ for some $a \in K$. (assumptions: $\text{char} K \neq 3$, $x^3 - a$ splits) Now look at when $\text{char} K = 3$. Then $x^3 - 1 = (x - 1)^3$, so only eigenvalue of σ is 1. Only eigenvectors in L are the elements of K (fixed by σ). So L is not spanned by its eigenvectors. How do we construct L ? Answer: Pick a generalized eigenvector. (LINEAR ALGEBRA REVIEW: an eigenvector is v with $(\sigma - \lambda)v = 0$, where λ is an eigenvalue and σ is a linear map. A generalized eigenvector is v , with $(\sigma - \lambda)^n v = 0$. If σ is a linear map of a finite-dim vector space V , then every element of V is a sum of generalized eigenvectors.) Now we have $K \subset L$, Galois group is $\mathbb{Z}/3\mathbb{Z}$. Pick α to be a generalized eigenvector of σ . We have $(\sigma - 1)^2\alpha = 0$, $(\sigma - 1)\alpha \neq 0$. Note that $(\sigma - 1)\alpha$ is fixed by σ in K . $\sigma\alpha = \alpha + a$ for some $a \in K$. Rescale α (can assume $\sigma\alpha = \alpha + 1$). Then $\sigma(\alpha^p - \alpha) = (\alpha + 1)^p - (\alpha + 1) = \alpha^p - \alpha$. So $\alpha^p - \alpha \in K$, fixed by σ . So $\alpha^p - \alpha - b = 0$ for some $b \in K$.

35. **Generalized eigenvector example 1.** Let $\sigma = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$. Then $\sigma^2 = 0$, so $(\sigma - 0)^2 v = 0$, so every vector is a generalized eigenvector. Eigenvectors are $\begin{pmatrix} x \\ 0 \end{pmatrix}$.

36. **Generalized eigenvector example 2.** Let σ be a Jordan block.

37. **Polynomial of degree ≤ 4 is solvable by radicals.** Here we show that a polynomial of degree ≤ 4 is solvable by radicals. Polynomials of degree ≥ 5 are usually not solvable, as we can take $x^5 - 4x + 2$ has Galois group S_5 , which is not a solvable group. However, $x^5 - 2$ is solvable. Now we show: if Galois group is solvable, then equation is solvable by radicals. To show this, take $1 = G_0 \subset G_1 \subset \dots \subset G_n$. Assume G_{i+1} (G_i is cyclic of the form $\mathbb{Z}/p_i\mathbb{Z}$ (p_i prime)). Add in all roots of unity - we need $1^{1/n}$. Then we get a chain of fields $L_0 \supset L_1 \supset L_2 \supset \dots \supset L_n$, where extension (like $L_1 \supset L_0$) is Galois, with Galois group $\mathbb{Z}/p_i\mathbb{Z}$. So, by the above results, $L_i = L_{i+1}(*^{1/p})$. So L_0 is solvable by radicals. Now, we show: equation solvable radicals implies Galois group is solvable. Take a sequence $L_n \subset \dots \subset L_{i+1} \subset L_i$, where $L_i = L_{i+1}(\sqrt{*})$. We have the extension $L_n \subset \dots \subset L_{i+1}$ is normal, and the extension $L_{i+1}(\sqrt{*})/L_i$ is also normal, but the extension $L_n \subset \dots \subset L_i$ might NOT be normal. We have $L_i = L_{i+1}(\sqrt{a}, \sqrt{a'}, \dots)$, where $a, a', \dots$ are all CONJUGATES with respect to the Galois group. Otherwise, we just reverse the above argument and the result holds.

38. **Factorization of $x^n - 1$ in $\mathbb{Z}[x]$.** Note that $x^n - 1 = (x - 1)(1 + x + \dots + x^{n-1})$. For the polynomials $x^1 - 1, x^2 - 1, x^3 - 1, x^4 - 1, x^5 - 1, x^6 - 1$, we get their respective factorizations to be $x - 1, (x - 1)(x + 1), (x - 1)(x^2 + x + 1), (x - 1)(x + 1)(x^2 + 1), (x - 1)(x^4 + x^3 + x^2 + x + 1), (x - 1)(x + 1)(x^2 - x + 1)(x^2 + x + 1)$. Note that 3rd root of 1 is also a 6th root of 1, so we thus get some factors in the

factorizations to be repeated. Looking at the definition of the n th cyclotomic polynomial, we get an obvious property that $x^n - 1 = \prod_{d|n} \Phi_d(x)$, where $\Phi_d(x)$ is the d th cyclotomic polynomial. We get a Mobius inversion formula $\Phi_n(x) = \prod_{d|n} (x^d - 1)^{\mu(n/d)}$, where μ is the Mobius function.

39. **Example of using Mobius inversion formula.** Using the Mobius inversion formula, we may calculate

$$\Phi_{30}(x) = \frac{(x^{30} - 1)(x^5 - 1)(x^3 - 1)(x^2 - 1)}{(x^{15} - 1)(x^{10} - 1)(x^6 - 1)(x - 1)}$$

40. **Values of cyclotomic polynomials.** We calculate the n th cyclotomic polynomials up to $n = 10$. These are, respectively, $x - 1, x + 1, x^2 + x + 1, x^2 + 1, x^4 + x^3 + x^2 + x + 1, x^2 - x + 1, x^6 + x^5 + x^4 + x^3 + x^2 + x + 1, x^4 + 1, x^6 + x^3 + 1, x^4 - x^3 + x^2 - x + 1$. Note that the automorphism $x \mapsto -x$ allows us to go from the $n = 5$ cyclotomic polynomial to the $n = 10$ cyclotomic polynomial. Similarly, the automorphism $x \mapsto x^3$ allows us to go from the 3rd cyclotomic polynomial to the 9th cyclotomic polynomial. Also notice that $\Phi_{27}(x) = x^{18} + x^9 + 1$. Primitive 10th roots of 1 are $(-1)(\text{primitive 5th roots of 1})$, same for $\Phi_{2 \times (\text{odd})}(x)$.

41. **Coefficients of cyclotomic polynomials.** Observe that all coefficients of cyclotomic polynomials seem to be $0, \pm 1$. Actually, this hypothesis fails, but the smallest case where a coefficient is not $0, \pm 1$ happens for the n th cyclotomic polynomial, with $n > 100$.

42. **Example of theorem that cyclotomic polynomials are irreducible but irred over finite fields.** For Φ_p (p prime), we have $\frac{x^p - 1}{x - 1}$. The map $x \mapsto x + 1$ gives $\frac{(x+1)^p - 1}{x}$. This satisfies Eisenstein's criterion. Note also that $p \mid (p \text{ choose } i)$ for $0 < i < p$. For Φ_{p^n} , same trick works. We have $\frac{x^{p^n} - 1}{x - 1} \mapsto \frac{(x+1)^{p^n} - 1}{x - 1}$. We also have that $\Phi_{2 \cdot p^n}(x) = \Phi_{p^n}(-x)$ (for p odd). These cases cover $n < 12$.

43. **Application of Φ_n .** There exists infinitely many primes $\equiv 1 \pmod{n}$. For $n = p$ prime, use polynomial $\frac{x^p - 1}{x - 1} = x^{p-1} + x^{p-2} + \dots + 1 = \Phi_p(x)$. For general n , proof is similar, but instead we use $\Phi_n(x)$ instead of $\frac{x^p - 1}{x - 1}$. Pick a prime q dividing $\Phi_n(x)$, and assume $(q, n) = 1$. All roots of $x^n - 1$ are distinct mod q , by assumption that q, n are coprime. So, integer a has order exactly n in $\mathbb{Z}/q\mathbb{Z}$. So $q \equiv 1 \pmod{n}$ (by Lagrange's theorem, that order of element divides order of group, where group is $(\mathbb{Z}/q\mathbb{Z})^*$, which has order $q - 1$). So if we have primes $p_1, \dots, p_k \equiv 1 \pmod{n}$, then put $p_1 \cdots p_k \cdot n$. Take $q \mid \Phi_n(a)$. Then $q \equiv 1 \pmod{n}$, with $q \neq p_1, \dots, p_k$, so we get infinitely many primes $\equiv 1 \pmod{n}$.

44. **Another application of Φ_n .** Given any abelian group G , there is a Galois extension of $\mathbb{Q}$ with Galois group G . To prove this, if G is a quotient of $(\mathbb{Z}/n\mathbb{Z})^*$ for some n , then the corresponding extension of $\mathbb{Q}$ has Galois group G . Note that $(\mathbb{Z}/n\mathbb{Z})^*$ is the Galois group of $x^n - 1$. So, just need to show that any abelian group is a quotient of some $(\mathbb{Z}/n\mathbb{Z})^*$. For an abelian group, we have (by structure theorem) that it is $\mathbb{Z}/n_1\mathbb{Z} \times \mathbb{Z}/n_2\mathbb{Z} \times \dots$ (pick distinct). For primes,

we have (distinct) $p_1 \equiv 1 \pmod{n_1}, p_2 \equiv 1 \pmod{n_2}, \dots$. Put $n = p_1 \cdot p_2 \cdots$. Then $(\mathbb{Z}/n\mathbb{Z})^* = \mathbb{Z}/(p_1 - 1)\mathbb{Z} \times \mathbb{Z}/(p_2 - 1)\mathbb{Z} \times \dots$. We thus get onto maps $\mathbb{Z}/(p_1 - 1)\mathbb{Z} \rightarrow \mathbb{Z}/n_1\mathbb{Z}$, and similarly for the others.

45. **Why Shavarevich's theorem is hard.** Recall that the theorem says that any solvable group is a Galois group over $\mathbb{Q}$. Suppose $1 \rightarrow A \rightarrow G \rightarrow B \rightarrow 1$ is an exact sequence (with G solvable). Pick $\mathbb{Q} \subset L$ (extension with Galois group B) with $L \subset M$. Can we extend this to a field M , so that $\text{Gal}(M/\mathbb{Q}) = G$? (not always possible) e.g. we could have $1 \rightarrow \mathbb{Z}/2\mathbb{Z} \rightarrow \mathbb{Z}/4\mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z} \rightarrow 1$. $\mathbb{Q} \subset \mathbb{Q}(\sqrt{n}) \subset M$. Can we find M with $\text{Gal}(M/\mathbb{Q}) = \mathbb{Z}/4\mathbb{Z}$? M is generated by a , with $\sigma^2 = -a$, where σ generates by $\mathbb{Z}/4\mathbb{Z}$. $\sigma(a)/a$ fixed by σ^2 , so is $\mathbb{Q}(\sqrt{n})$. So $\sigma(a)/a = b + c\sqrt{a}$, with $b, c \in \mathbb{Q}$. Then $b^2 - nc^2 = (b + c\sqrt{n})(b - \sqrt{n}) = \sigma(a)/a \cdot \sigma^2(a)/\sigma(a) = -1$. So $b^2 - nc^2 = -1$, but cannot solve this if $n = -1$. So $\mathbb{Q}[i]$ is not contained in any Galois extension with Galois group $\mathbb{Z}/4\mathbb{Z}$. Also fails if $n = 3, 7, 11, \dots$. In modulo 4, if $n \equiv 3 \pmod{4}$, then $b^2, c^2 \equiv 0, 1 \pmod{4}$, so $b^2 - nc^2 \equiv 0, 1, 2 \pmod{4}$. But $-1 \equiv 3 \pmod{4}$.

46. **Another application of Φ_n .** Every finite division ring is a field. So there are no finite analogues of the quaternions. Suppose L is a finite division ring. Center is a field, say F_q . L is a finite-dim vector space over F_q for some q . Say dimension is m . Then look at group L^* (the set of nonzero $l \in L$). Look at class equation. $q^m - 1 = q - 1 + \sum \frac{q^m - 1}{q^* - 1}$, where LHS is the number of elements of L^* , $q - 1$ is the conjugacy class with size 1 (the center), and $q^* - 1$ corresponds to centralizer of any element of subspace of L , so has size q^* for some number $*$. $\Phi_m(q)$ divides $q^m - 1$, $\frac{q^m - 1}{q^* - 1}$ when $*$ $\neq 0$. So $\Phi_m(q)$ divides $q - 1$, and this is only possible if $m = 1$, since $\Phi_m(q) = \prod_{\text{prim mth root of } 1}(q - \zeta)$, and $q - \zeta$ has absolute value $< q - 1$ (see diagram in notes). So $m = 1$, so $L = F_q$ is commutative.

47. **$2 \times 2 \times 2$ hyperdeterminants.** See humongous formula given in class.

48. **Norm, trace example for $\mathbb{R} \subset \mathbb{C}$.** Suppose we have the field extension $\mathbb{R} \subset \mathbb{C}$. Let it have basis $1, i$ over $\mathbb{R}$. Write $z = x + iy \in \mathbb{C}$. Look at z as a linear map, which gives the matrix $A = \begin{pmatrix} x & y \\ -y & x \end{pmatrix}$. Thus $\text{trace}(x + iy) = \text{trace}(A) = 2x = 2\text{Re}(z)$ and $\text{Norm}(x + iy) = \det(A) = x^2 + y^2 = |z|^2$.

49. **Calculating $N(\alpha), \text{Tr}(\alpha)$.** Suppose $L = K(\alpha)$. Look at minimal polynomial of $\alpha : \alpha^n + a_{n-1}\alpha^{n-1} + \dots + a_0 = 0$. What is norm, trace of α ? Basis of L is $1, \alpha, \alpha^2, \dots, \alpha^{n-1}$ (α generates L). In this basis, $\beta \mapsto \alpha\beta$. Looks like

$$\begin{pmatrix} 0 & 1 & 0 & 0 & \dots & 0 \\ 0 & 0 & 1 & 0 & \dots & 0 \\ \vdots & & & & & \\ 0 & 0 & 0 & 0 & \dots & 1 \\ -a_0 & & & & & -a_{n-1} \end{pmatrix}$$

where the first row corresponds to $\alpha \mapsto \alpha$, the second row corresponds to $\alpha \mapsto \alpha^2$, ..., and the last row corresponds to $\alpha^n = -a_0 - a_1\alpha - \dots$. So

$\text{trace}(\alpha) = -a_{n-1}$ and $\text{norm}(\alpha) = (-1)^n \cdot a_0$. In general, $\text{trace}(\alpha) = [L : K(\alpha)] \times (-a_{n-1})$ and $\text{norm}(\alpha) = ((-1)^n a_0)^{[L:K(\alpha)]}$. For a Galois extension, $\text{norm}(\alpha) = \prod_{g \in G} g(\alpha)$ and $\text{trace}(\alpha) = \sum_{g \in G} g(\alpha)$.

50. **Application of norm, trace.** We want to find the ring of integers of a quadratic field $\mathbb{Q}(\sqrt{n})$, i.e. we want something satisfying the equation $\alpha^n + a_{n-1}\alpha^{n-1} + \cdots + a_0 = 0$, where the leading coeff is 1 and the other coeffs are integers. For instance, the ring of integers of $\mathbb{Q}(\sqrt{2})$ is just $\mathbb{Z}(\sqrt{2})$. We have a similar situation for $\mathbb{Q}(\sqrt{3})$. But for $\mathbb{Q}(\sqrt{5})$, we have that $\phi = \frac{1+\sqrt{5}}{2}$ is an algebraic integer, as $\phi^2 = \phi + 1$. If α is an (algebraic) integer, then $N(\alpha), \text{Tr}(\alpha)$ are both integers. $\alpha^n + a_{n-1}\alpha^{n-1} + \cdots + a_0 = 0$. Suppose $\alpha = a + b\sqrt{n}$ is an algebraic integer. Then $N(\alpha), \text{Tr}(\alpha)$ are integers. Basis is $1, \sqrt{n}$. Matrix for α is $\begin{pmatrix} a & b \\ nb & a \end{pmatrix}$. So $\text{Tr}(\alpha) = 2a$ and $N(\alpha) = a^2 - nb^2$, where both of these are integers. As an example, take $\mathbb{Q}(\sqrt{2})$. For $a + b\sqrt{2}$, we have $2a \in \mathbb{Z}$, $a^2 - 2b^2 \in \mathbb{Z}$ (either $a \in \mathbb{Z}$ or $a \in \mathbb{Z} + 1/2$, we want to eliminate the latter case). We have that $2b + a\sqrt{2}$ is an algebraic integer. $2(2b) \in \mathbb{Z}$. If $a = 1/2$ (say), then $(1/2)^2 - 2b^2 = n$, so $1 - 8b^2 = 4n$, $4b \in \mathbb{Z}$. $2 - 16b^2 = 8n$, $2 - c^2 = 8n$, $c^2 \equiv 2 \pmod{8}$, which is a contradiction. So, in fact, $a \in \mathbb{Z}$, NOT $a \in \mathbb{Z} + 1/2$. Easy to check that then $b \in \mathbb{Z}$, so algebraic integers are just $a + b\sqrt{2}$, with $a, b \in \mathbb{Z}$. For $\mathbb{Q}(\sqrt{5})$, when is $a + b\sqrt{5}$ an algebraic integer? It is known that $2a \in \mathbb{Z}$. It is possible for $a \in \mathbb{Z} + 1/2$. Look at $\frac{1+\sqrt{5}}{2}$. All algebraic integers are of the form $m + \frac{n\sqrt{5}+1}{2}$ (proof omitted), for $m, n \in \mathbb{Z}$. Subtract multiple of $\frac{1+\sqrt{5}}{2}$, and can assume $a = 0$. $b\sqrt{5}$ algebraic integer implies that $b \in \mathbb{Z}$, and $x^2 - 5b^2 = 0$.

51. **Application of Artin's lemma.** Suppose $K \subset L$ is a finite extension. We have trace map $\text{Tr} : L \rightarrow K$. Is trace nonzero? Yes, obvious if $\text{char} = 0$, as then $\text{Tr}(1) = [L : K] \neq 0$. Also look at $F_p(t) \subset F_p(t^{1/p})$ (trace is always zero). Now we show that trace is nonzero for separable extensions, we prove it for Galois extensions. We have $\text{Tr}(x) = \sum_i \sigma_i(x)$, where $\sigma_i \in \text{Galois group}$. Note that $\sum_i \sigma_i(x)$ is a linear combination of $\sigma_1, \sigma_2, \dots$. So it is nonzero for some x by Artin's lemma. So $\text{Tr}(x) \neq 0$ for some x . Now we look at a symmetric bilinear form on L , given by $(\alpha, \beta) = \text{Tr}(\alpha\beta)$. If L/K is separable, then $(\cdot, \cdot)$ is nondegenerate. WARNING: if L/K is a quadratic extension, we have a quadratic form $N(\alpha)$ and a quadratic form $\text{Tr}(\alpha \cdot \alpha)$. BUT THESE QUADRATIC FORMS ARE DIFFERENT! (even for $\mathbb{R} \subset \mathbb{C}$) In the extension $\mathbb{R} \subset \mathbb{C}$, we have $N(x + iy) = (x + iy)(x - iy) = x^2 + y^2 = |x + iy|^2$ and $\text{Tr}((x + iy)(x + iy)) = \text{Tr}(x^2 + 2ixy - y^2) = 2(x^2 - y^2)$, but note that the first quadratic form is positive definite but the second quadratic form is indefinite.

52. **Discriminant of a quadratic form.** Choose a basis so the form is diagonal. Write $\sum a_i x_i^2$. Discriminant is $\prod a_i$. Seems not well-defined. Only defined as an element of $L^*/(L^*)^2$, where the denominator is squares of L . Suppose the quadratic form is $\sum a_{ij} x_i x_j$ ($a_{ij} = a_{ji}$ and $A = a_{ij}$). Then discriminant is $\det(a_{ij})$ (= product, if diagonal). Look at change of basis using B . Then change A to BAB^t (which is NOT equal to BAB^{-1} , as the latter is a change

of basis of linear maps). Thus $\det(BAB^t) = \det(B)^2 \cdot \det(A)$, so determinant changes by a square, so $\det(A)$ is well-defined in $L^*/(L^*)^2$.

53. **Calculating discriminant example.** Let L_1 = extension of $\mathbb{Q}$, given by root of $x^3 + x + 1$. Let L_2 be defined similarly but for $x^3 - x + 1$. Want to show that these extensions have DIFFERENT discriminants. How to calculate discriminants? Suppose L_1/K is a field extension, generated by α . Look at matrix

$$\begin{pmatrix} \sigma_1(1) & \sigma_1(\alpha) & \sigma_1(\alpha^2) \\ \sigma_2(1) & \sigma_2(\alpha) & \dots \\ \vdots & & \end{pmatrix}$$

(where $\sigma_1(\alpha), \sigma_2(\alpha), \dots$ are all conjugates. The determinant of the above matrix is $\prod_{i < j} (\sigma_i(\alpha) - \sigma_j(\alpha))$ (Vandermonde determinant). Multiplying the above matrix by its transpose gives

$$\begin{pmatrix} Tr(1 \cdot 1) & Tr(1 \cdot \alpha) & Tr(1 \cdot \alpha^2) \\ Tr(\alpha \cdot 1) & & \\ & & Tr(\alpha^i \cdot \alpha^j) \end{pmatrix}$$

since $Tr(x) = \sum_i \sigma_i(x)$. The determinant of this matrix is just discriminant calculated with basis $1, \alpha, \alpha^2, \dots, \alpha^{n-1}$. NOTE that $\prod_{i < j} (\sigma_i(\alpha) - \sigma_j(\alpha)) = \sqrt{\text{discriminant of polynomial}}$. So $\det(Tr(\alpha\beta))$ is discriminant of field generated by root of poly $f(x)$, which equals discriminant of $f(x)$ (EQUAL UP TO SQUARES), and the discriminant of $f(x)$ is equal to $\prod (\alpha_i - \alpha_j)^2$, where α_i are roots of f . For $f = x^3 + bx + c$, the discriminant is $-4b^3 - 27c^2$, so finding the discriminants of the polys $x^3 - x + 1$ and $x^3 + x + 1$ gives -27 and -31 , respectively. Note the ratio $-27/-31$ is NOT a square, so the fields are DIFFERENT.

54. **Application of Artin's lemma / Hilbert's Theorem 90.** Suppose $K \subset L$ with norm map $N : L^* \rightarrow K^*$. Which elements have norm 1? (Analogue of unit circle in $\mathbb{C} \rightarrow \mathbb{R}$, where $N(z) = 1 \iff |z| = 1 \iff z$ lies in the unit circle in $\mathbb{C}$) What is the analogue of unit circle for an algebraic number field? Look at finite fields first. Is $N : L^* \rightarrow K^*$ onto? NO! Look at $\mathbb{C} \rightarrow \mathbb{R}$ with $x + iy \mapsto x^2 + y^2$, but the image of $\mathbb{C}^* \rightarrow \mathbb{R}^*$ only lies in the positive real numbers. But it IS onto for finite fields. Look at $N(x) = \prod_{\sigma \in Gal} \sigma(x)$. Galois group is cyclic of order n generated by $x \mapsto x^q$ ($|K| = q$, $|L| = q^n$). So $N(x) = F^0(x)F^1(x) \cdots F^{n-1}(x)$ (where $F : x \mapsto x^q$ is the Frobenius map) $= x \cdot x^q \cdot x^{q^2} \cdots x^{q^{n-1}} = x^{1+q+\cdots+q^{n-1}}$. When is $|N(x)| = 1$? Number of solutions to $1 = x^{1+q+\cdots+q^{n-1}}$ is at most $q^{n-1} + \cdots + q + 1$ (as roots of $x^{1+q+\cdots+q^{n-1}} - 1 = 0$). So get a map $1 \rightarrow (\text{elements of norm 1}) \rightarrow L^* \xrightarrow{N} K^*$, where (elements of norm 1) has order at most $1 + q + \cdots + q^{n-1}$, L^* has order $q^n - 1$, and K^* has order $q - 1$. Key point is that $q^n - 1 = (q - 1)(1 + q + \cdots + q^{n-1})$. So $\{\text{norm 1 elements}\}$ has order $1 + q + \cdots + q^{n-1}$. So image has order $q - 1$. So the map $L^* \xrightarrow{N} K^*$ is onto. The map $1 \rightarrow (\text{elements of norm 1}) \mapsto L^* \xrightarrow{N} K^*$ is similar to the polar decomposition see in $1 \rightarrow (\text{unit disc}) \rightarrow \mathbb{C}^* \rightarrow \mathbb{R}_+^*$.

55. **Norm example, obvious case.** Obvious case when $N(x) = 1$. ($x = y/y^q$, conjugate of y under $y \mapsto y^q$). Note that y, y^q have the same norm. $y, \sigma y$ have same norm for any element σ of the Galois group, since $N(y) = \prod_{\tau} \tau(y) = \prod_{\tau} \tau\sigma(y)$, so $N(x/\sigma(y)) = 1$.
56. **Application of Hilbert's theorem 90.** Suppose $K \subset L$ is Galois, cyclic, and suppose K contains primitive n th roots of unity (write z as a prim. n th root of unity). Then $N(z) = z \cdots z$ (n times). Hilbert's theorem 90 then implies that $z = \alpha/\sigma\alpha$ for some $\alpha \in L^*$. So $\sigma\alpha = z\alpha$, so $\alpha^n \in K$, $L = K(\alpha^{1/n})$.
57. **Cohomology example.** Suppose G acts on $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$. Easy to check: $0 \rightarrow H^0(G, A) \rightarrow H^0(G, B) \rightarrow H^0(G, C) \rightarrow 0$ (exact). If $G = \mathbb{Z}/2\mathbb{Z}$, then we suppose $0 \rightarrow \mathbb{Z} \rightarrow \mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z} \rightarrow 0$, where G acts as -1 on the $\mathbb{Z}$'s. Then $H^0 : 0 \rightarrow 0 \rightarrow 0 \xrightarrow{NOTONTO} \mathbb{Z}/2\mathbb{Z}$. Now we give motivation for $H^1(G, M)$. We have an exact sequence $0 \rightarrow H^0(G, A) \rightarrow H^0(G, B) \rightarrow H^0(G, C) \rightarrow H^1(G, A) \rightarrow H^1(G, B) \rightarrow H^1(G, C)$. Definition of $H^1(G, M) = \frac{1\text{-cocycles}}{1\text{-coboundaries}}$. 1-cocycle has that $\sigma \mapsto a_{\sigma}$ and $a_{\sigma\tau} = a_{\sigma} + \sigma a_{\tau}$. 1-coboundary has that $a_{\sigma} = \sigma b - b$ for some b . It can be shown that $H^1(G, M)$ is a group and that we get exactness.
58. **Another cohomology example.** Take a cyclic group G , and let it act on L^* . Note that $a_{\sigma\tau} = a_{\sigma} \cdot \sigma(a_{\tau})$. Elements of Galois group G are $1, \sigma, \sigma^2, \dots$ as we assume σ generates G , as G is cyclic. We want elements $a_1, a_{\sigma}, a_{\sigma^2}, \dots$. We have $a_{\sigma^2} = a_{\sigma} \cdot \sigma a_{\sigma} = x \cdot \sigma x$ ($a_{\sigma} = x$). Then $a_{\sigma^3} = a_{\sigma^2} \cdot \sigma^2 a_{\sigma} = x \cdot \sigma x \cdot \sigma^2 x \dots$. Then $a_{\sigma^n} = a_1 = x \sigma x \cdots \sigma^n(x) = N(x)$. 1-coboundaries: $\alpha/\sigma\alpha$. $H^1(G, L^*) = \frac{\ker(N: L^* \rightarrow K^*)}{\alpha/\sigma\alpha}$, and this 1st cohomology group gives a cyclic version of Hilbert's theorem 90.